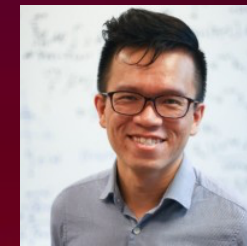




Giovanni Vignale
NUS



Justin Song
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Superconducting Berry curvature dipole

Oles Matsyshyn

NTU Singapore

Quantum Geometry in Quantum Materials

References:

Superconducting Berry curvature dipole

Oles Matsyshyn, Giovanni Vignale, **JS**, arXiv 2410.21363 (2024)

Superconducting Photocurrent and Light-Enriched Supercurrent-Phase Relation

Oles Matsyshyn, **JS**, arXiv 2507.06290 (2025)

Funding:



Ministry of Education
SINGAPORE

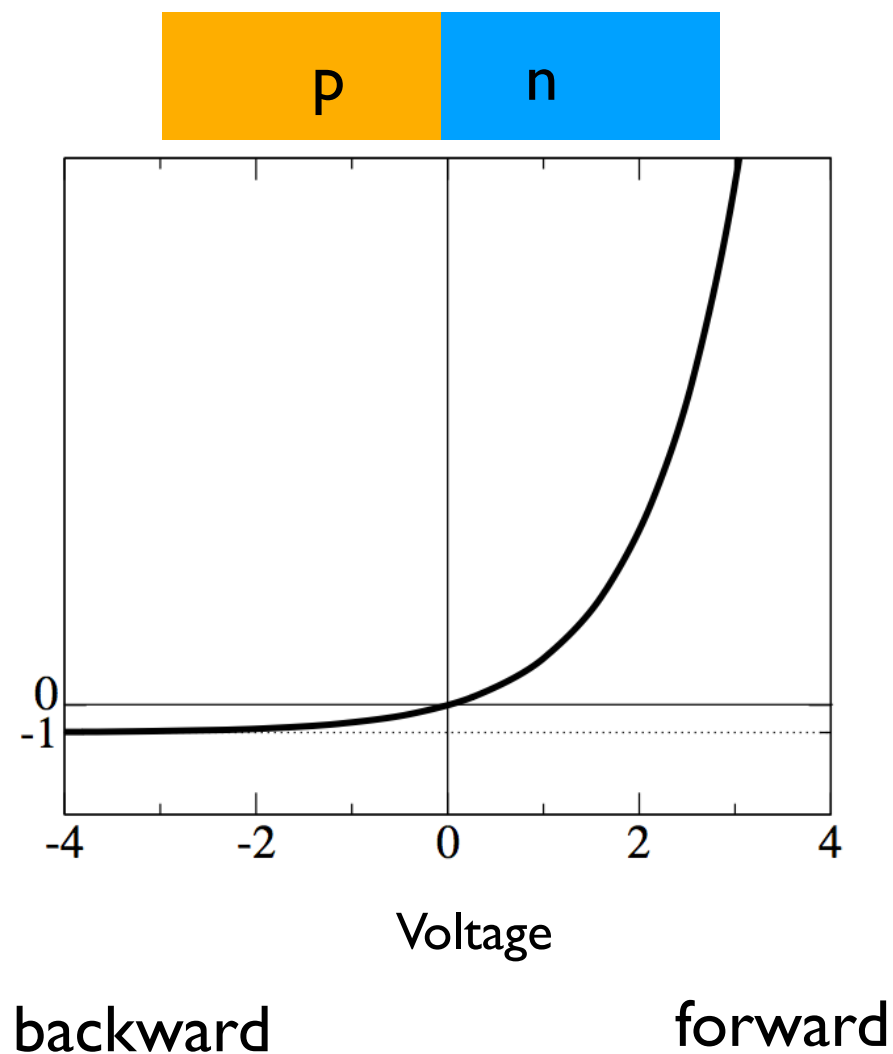
Main hero of today's talk is the Berry curvature dipole in superconducting material
In the field of optics Berry curvature dipoles are mostly known from the non-linear Hall response
So we start by clarifying what is the non-linear optical response
Under optical response I imply the response to external electric fields, what type? Can be different
Magnetisation, current, polarisation
Here we talk about current

“Classic” nonlinearity in a semiconducting device

For non-reciprocal devices

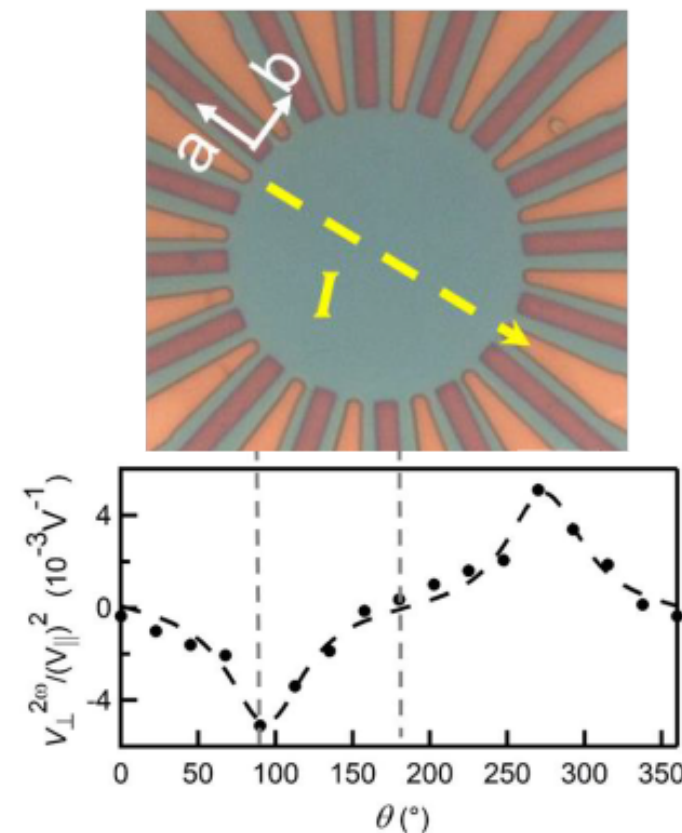
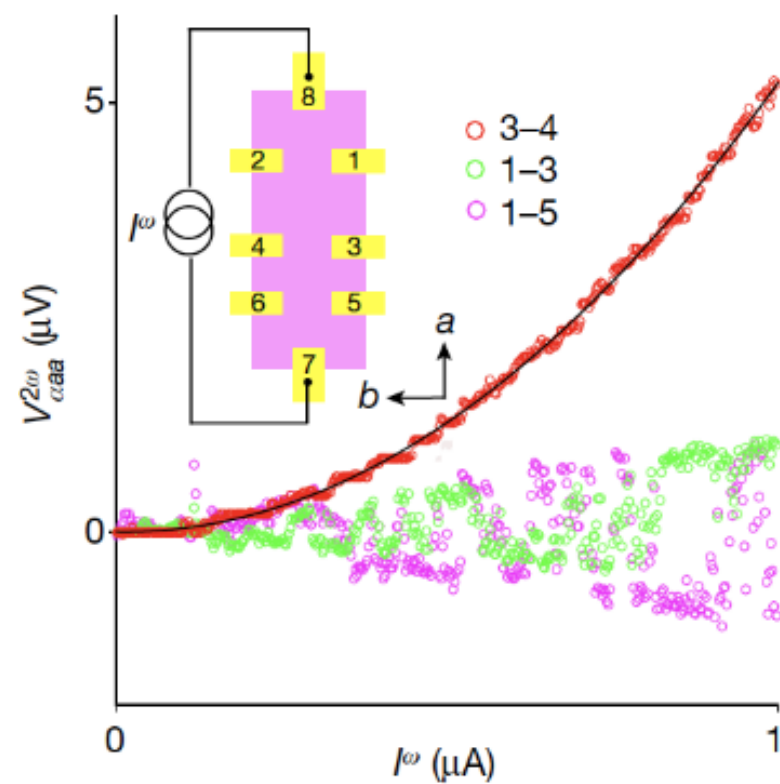
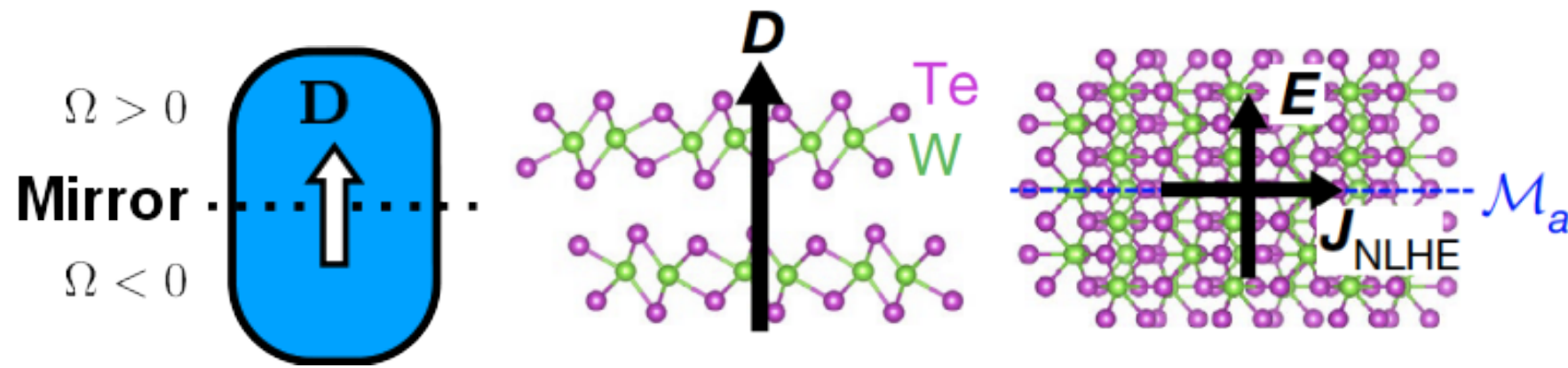
Static applied Electric field vs current

Requires **non-centrosymmetric** crystal structure (e.g., BBO, KTP, LiNbO₃)



Nonlinear responses without p-n junction

Nonlinear Hall response in a TRS preserving system



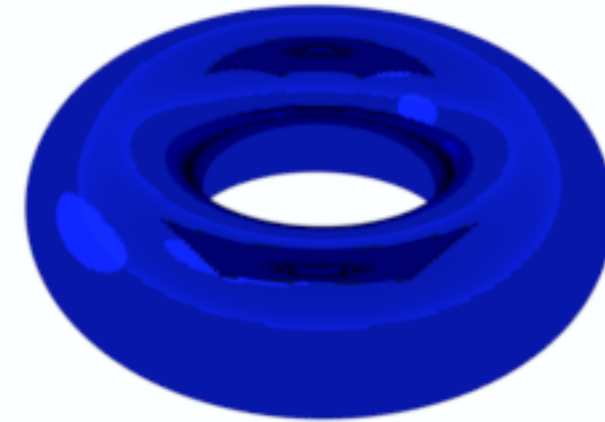
Kang et al., Nature Materials 18, 324-328 (2019)

Ma, et al., Nature 565, 337-342 (2019)

*see also excellent articles by Sodemann, Fu PRL (2015), and Morimoto, Nagaosa Science Advances (2016)

Linear Hall effect

Electrons on a lattice, when coupled to electromagnetic field develop anomalous group velocity



Classic velocity

Anomalous velocity

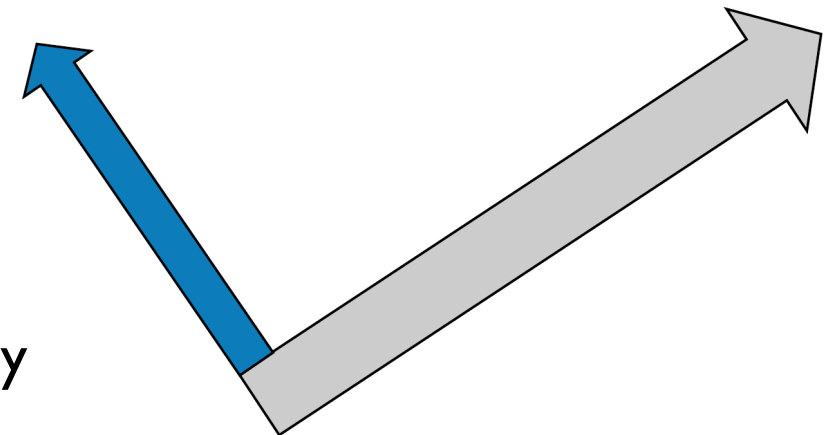
$$v_n^\alpha = \hbar^{-1} \partial_{\mathbf{k}} \epsilon_n(\mathbf{k}) - e E^\beta \Omega_n^{\alpha\beta}$$

Berry curvature

Controlled by quantum geometry
of the Bloch states

$\mathbf{E}$

$\mathbf{v}_{\text{anomalous}}$



$$\mathbf{j} = e \sum_{\mathbf{k}} \mathbf{v}_{\mathbf{k}} f(\mathbf{k})$$

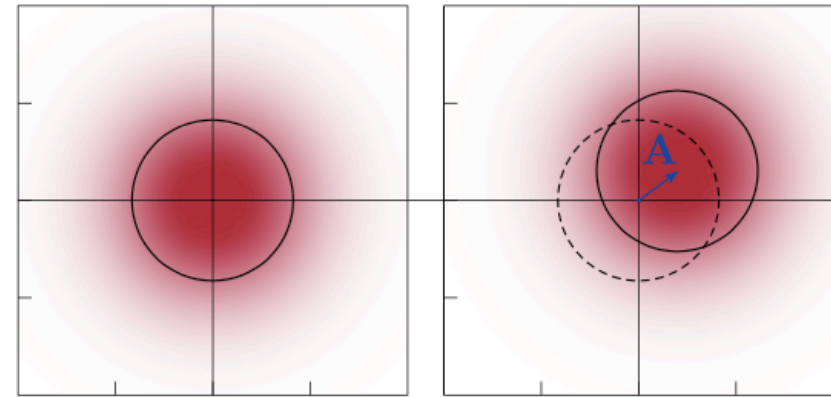
$$\text{TRS} : \Omega_{\mathbf{k}} = -\Omega_{-\mathbf{k}}$$

Net linear Hall response requires TRS breaking

Electronic second-order nonlinearities

current

$$\mathbf{j} = e \sum_{\mathbf{k}} \mathbf{v}_{\mathbf{k}} f(\mathbf{k})$$



in the presence of E field, occupation is displaced, by expanding in E we find:

$$f(\mathbf{k}) = f(\mathbf{k}) + f^{(1)}(\mathbf{k}) + f^{(2)}(\mathbf{k}) + \dots$$

group velocity

$$v_n^\alpha = \hbar^{-1} \partial_{\mathbf{k}} \epsilon_n(\mathbf{k}) - e E^\beta \Omega_n^{\alpha\beta}$$

in TRS materials, group velocity:

$$\mathbf{v}(\mathbf{k}) = -\mathbf{v}(-\mathbf{k})$$

Solving simple kinetic equation:

$$f^{(1)} \sim \mathbf{E} \partial_{\mathbf{k}} f(\mathbf{k})$$

in TRS materials, first order correction:

$$f^{(1)}(\mathbf{k}) = -f^{(1)}(-\mathbf{k})$$

Non-linear Hall effect survives! Requires Fermi surface.

Manifold of non-linear responses

velocity can be altered in the presence
of spatially varying electric fields

$$\tilde{\epsilon}(\mathbf{k}) \approx \epsilon(\mathbf{k}) + \frac{e}{2} \nabla_{\mathbf{r}} \cdot [g(\mathbf{k}) \mathbf{E}(\mathbf{r})]$$

Hughes and Lappa PRB (2019)

Berry curvature can even be altered in
the presence of electric fields

$$\bar{\Omega}_s(\mathbf{k}) = \Omega_s(\mathbf{k}) + \nabla_{\mathbf{k}} \times \mathcal{G}(\mathbf{k}) \mathcal{E}(t).$$

Gao, Yang Niu, PRL (2014)

associated with “Chiral magnetic effect”:
current induced by a magnetic field
when electrons of different chirality are
imbalanced.

$$\begin{aligned} \dot{\mathbf{r}} &= \left(1 + \frac{e}{c} \mathbf{B} \cdot \boldsymbol{\Omega}_{\mathbf{p}}\right)^{-1} \left[\mathbf{v} + e \mathbf{E} \times \boldsymbol{\Omega}_{\mathbf{p}} + \frac{e}{c} (\boldsymbol{\Omega}_{\mathbf{p}} \cdot \mathbf{v}) \mathbf{B} \right], \\ \dot{\mathbf{p}} &= \left(1 + \frac{e}{c} \mathbf{B} \cdot \boldsymbol{\Omega}_{\mathbf{p}}\right)^{-1} \left[e \mathbf{E} + \frac{e}{c} \mathbf{v} \times \mathbf{B} + \frac{e^2}{c} (\mathbf{E} \cdot \mathbf{B}) \boldsymbol{\Omega}_{\mathbf{p}} \right], \end{aligned}$$

Berry curvature and magnetic field alter density of states

$$n_e = \int^{\mu} \frac{d\mathbf{k}}{(2\pi)^d} \left(1 + \frac{e \mathbf{B} \cdot \boldsymbol{\Omega}}{\hbar}\right),$$

Xiao, Shi, Niu PRL 2005

associated with “Chiral anomaly”:
pumping of charge between particles
of different chirality

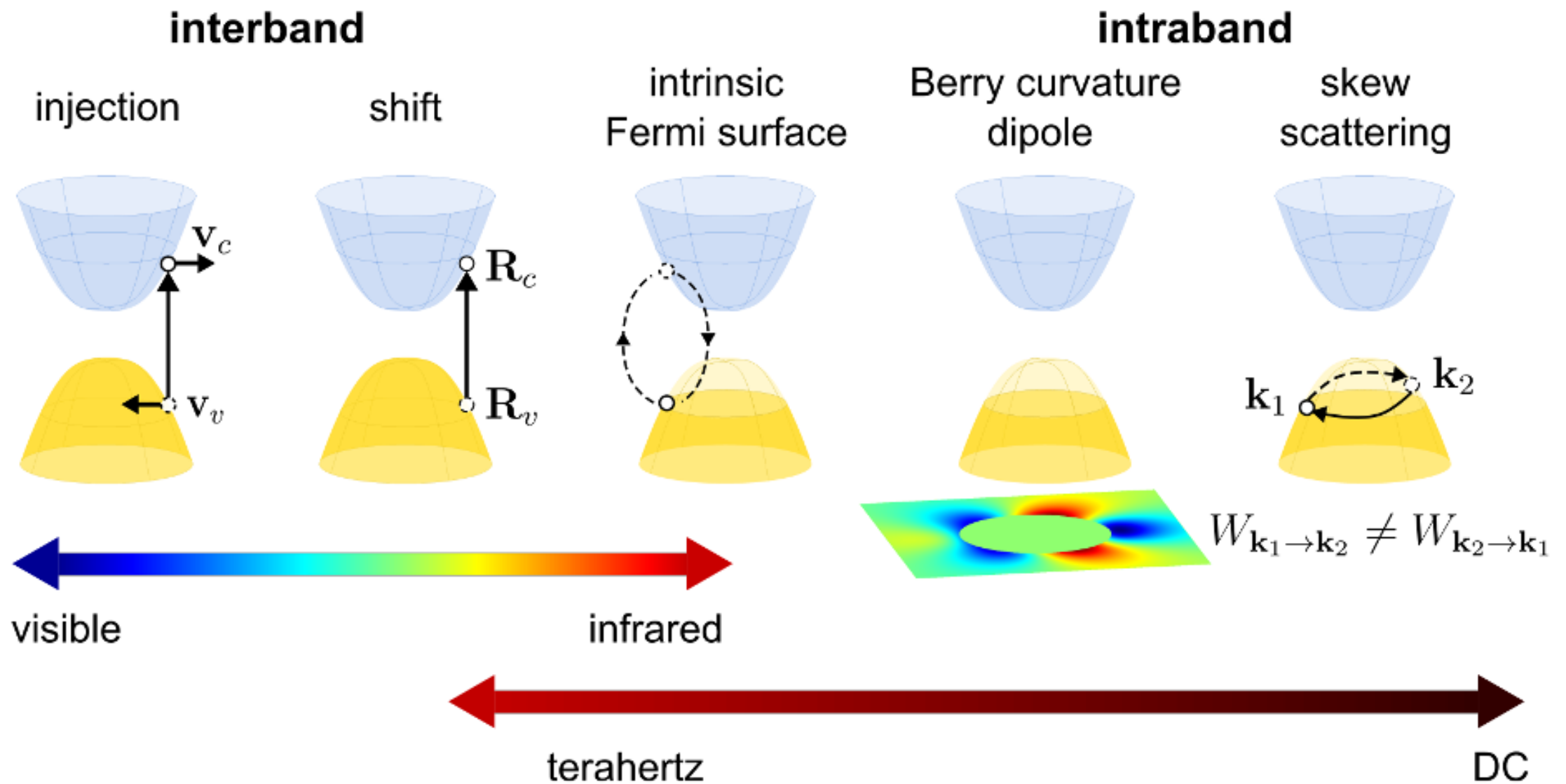
Son and Spivak Phys. Rev. B **88**, 104412 (2013)

Note: All of these follow from the full, non-linear Liouville quantum kinetic equation.

Why BCD is so interesting?

For TRS dominates low frequency response

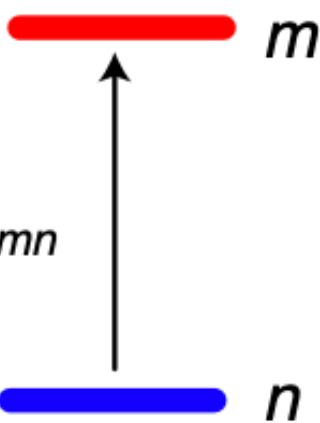
$$\mathbf{j}_{\text{BCD}}(\omega) \sim 1/\omega$$



Y Xiong, O Matsyshyn, RK Kumar, JCW Song, chapter quantum geometric opto-electronics as part of "Roadmap for 2D Photonics" (ACS Photonics to appear)

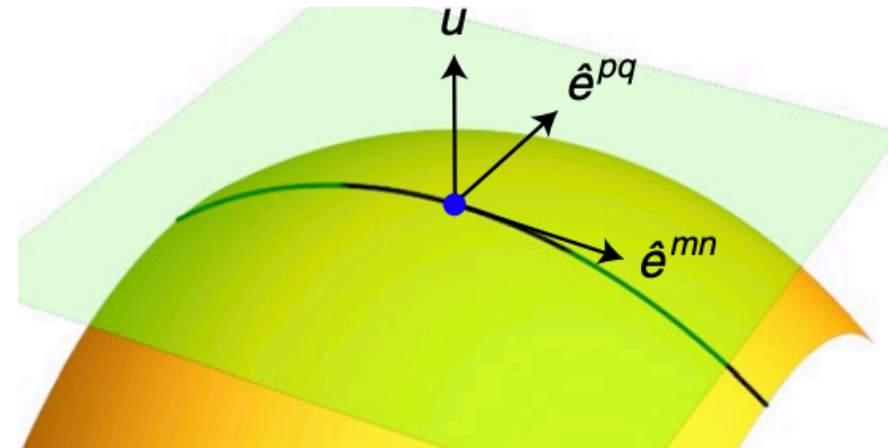
Noninteracting interband quantum geometry in **k**

interband transition dipole matrix element

$$\langle \psi_m | \hat{H}_{\text{int}} | \psi_n \rangle = e \mathbf{E} \cdot \mathbf{r}_{mn}$$


transition dipole moment matrix elements

$$\hat{e}_a^{mn}(\mathbf{k}) \equiv r_{mn}^a(\mathbf{k}) |u_{m\mathbf{k}}\rangle \langle u_{n\mathbf{k}}|$$



interband quantum geometry governs interband optical responses

e.g.,

Quantum geometric tensor

$$Q_{ba}^{mn} \equiv (\hat{e}_b^{mn}, \hat{e}_a^{mn}) = r_{nm}^b r_{mn}^a.$$



Optical absorption

$$\sigma^{b;a} = \frac{\pi \omega e^2}{h} \sum_{m,n} \int_{\mathbf{k}} \delta(\omega - \omega_{mn}) f_{nm} Q_{ba}^{mn},$$

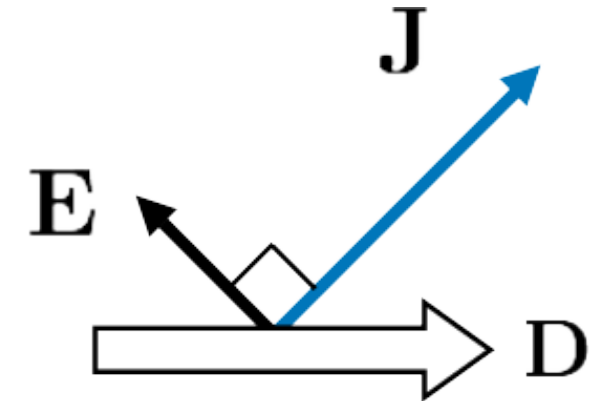
Why BCD is so interesting?

For TRS dominates low frequency response

$$\mathbf{j}_{\text{BCD}}(\omega) \sim 1/\omega$$

Doesn't do work!

$$\mathbf{j} \cdot \mathbf{E} = 0$$



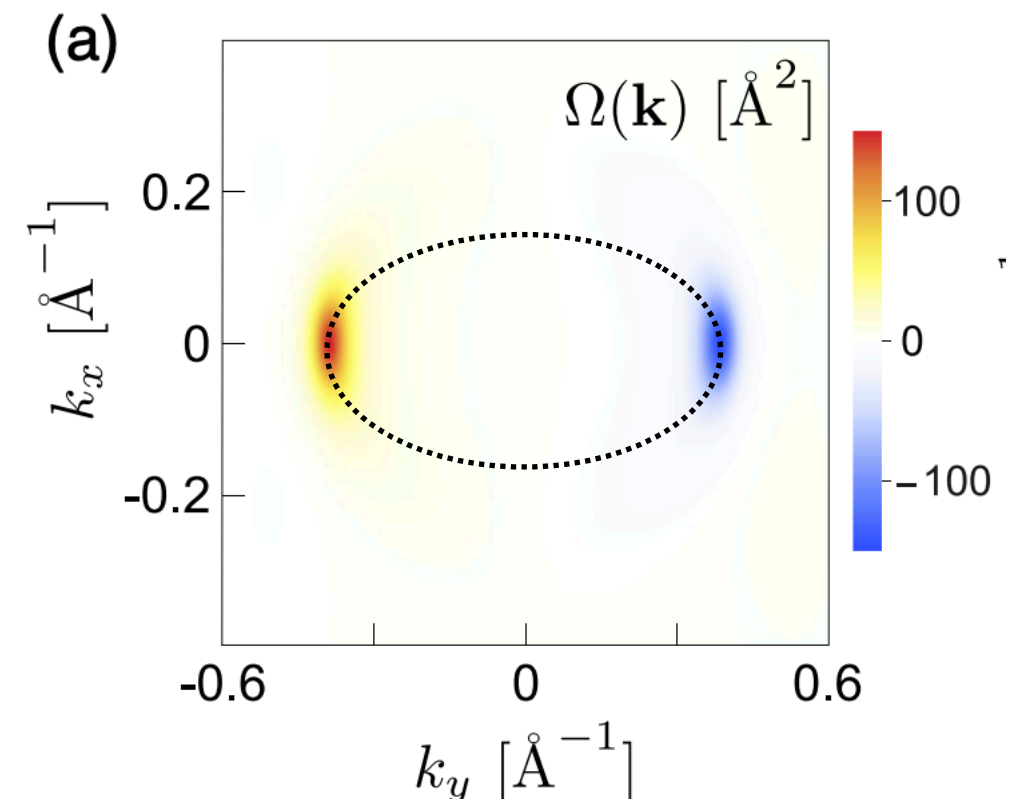
Directly measures the Berry curvature dipole! (2D):

$$\mathbf{j} = e^3 \tau (\mathbf{D} \cdot \mathbf{E}) \hat{z} \times \mathbf{E}$$

where

$$\mathbf{D} \equiv \int \frac{d^d k}{(2\pi)^d} f_0(\mathbf{k}) \frac{\partial \Omega(\mathbf{k})}{\partial \mathbf{k}}$$

Berry curvature distribution in WTe2 monolayer

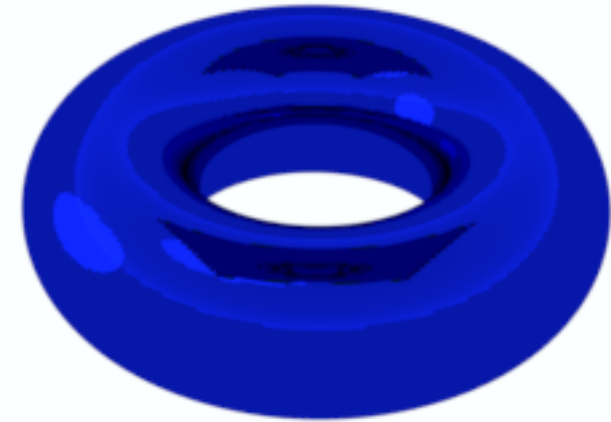


I. Sodemann and L. Fu, Phys. Rev. Lett. 115, 216806 (2015)

There are also other intrinsic NLHEs. e.g., the so-called “intrinsic” nonlinear Hall effect arising from E field corrections to the Berry curvature [see Y Gao, SA Yang, Q Niu, PRL 112, 166601 (2014), C Wang, Y Gao, D Xiao, PRL 277201 (2021), Liu et al. PRL 127, 277202 (2021)]

Linear Hall effect

Electrons on a lattice, when coupled to electromagnetic field develop anomalous group velocity



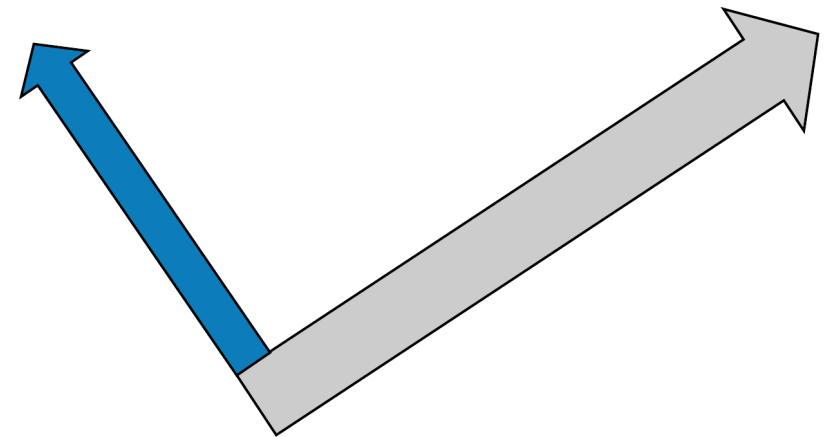
Non-linear Drude weight



$$a_n^\mu = -E^\gamma \partial^\gamma \partial^\mu \omega_n + E^\gamma E^\alpha \partial^\gamma \Omega_n^{\alpha\mu}$$

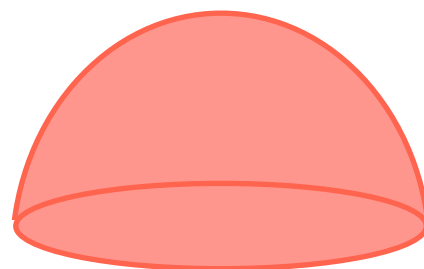
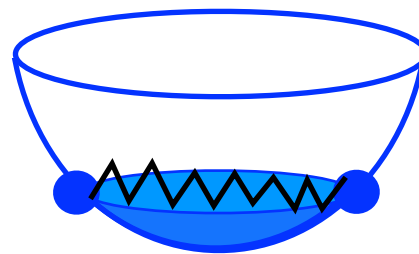
E

v_{anomalous}



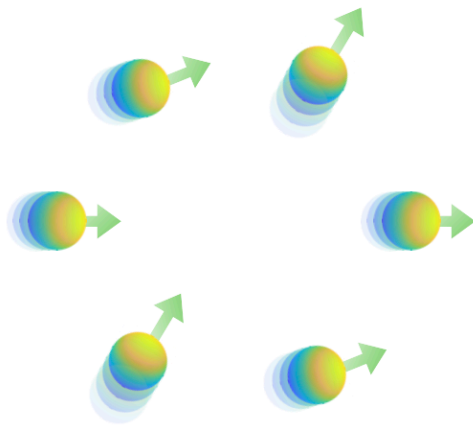
How do interactions change the Berry curvature dipole?

Cooper Pair



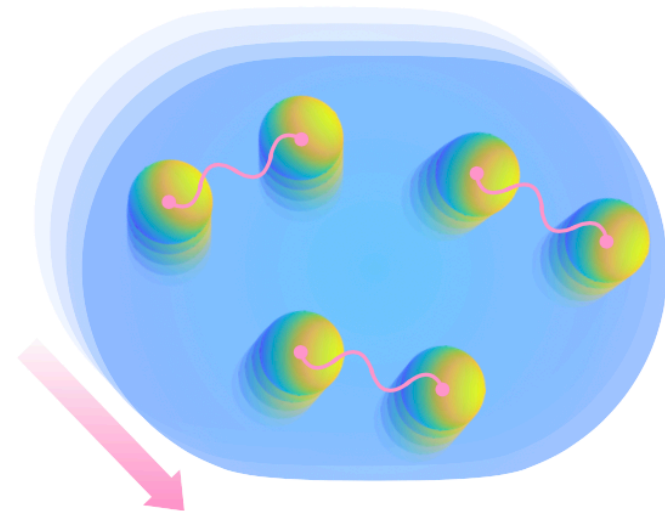
Qn: How does
superconductivity
change the Berry
curvature dipole?

Metal vs Superconductor



$$|\text{FS}\rangle = \prod_{\mathbf{k} \leq \mathbf{k}_F} c_{\mathbf{k}}^{\dagger} |0\rangle$$

- * gapless excitations
- * can be understood as filling particles up to Fermi surface
- * geometric quantities dominated by Fermi surface (Haldane PRL 2004)
- * Dynamics/responses such as the nonlinear Hall effects arise from motion of particles at the Fermi surface



$$|\text{SC}\rangle = \prod_{\mathbf{k}} (u_{\mathbf{k}} + v_{\mathbf{k}} c_{\mathbf{k},\uparrow}^{\dagger} c_{-\mathbf{k},\downarrow}^{\dagger}) |0\rangle$$

- * gapped charge excitations
- * Dynamics arise from coherent motion of **all** Cooper pairs
- * geometric quantities from all Cooper pairs
- * No Fermi surface! Does the Berry curvature dipole survive?

Flux threading characterizes many-body phases

Kohn “Theory of the Insulating State”

Threading flux (“vector potential”)

$$\hat{H}(\boldsymbol{\kappa}) = \frac{1}{2m} \sum_{i=1}^N (\hat{\mathbf{p}}_i + \hbar \boldsymbol{\kappa})^2 + \hat{V}_{\text{lattice}} + \hat{H}_{\text{int}},$$

↑
flux

Extracting the “Drude weight”

$$\lim_{\omega \rightarrow 0} \{ \omega \text{Im}[\sigma(\omega)] \} = -\frac{1}{\text{vol.}} \frac{d^2 E(\boldsymbol{\kappa})}{d\boldsymbol{\kappa}^2}$$

W Kohn, Physical Review 133, A171 (1964)

Key point: Energy of insulators insensitive to flux but energy of metal is

The BCD originated in how the electrons dislocated in the presence of A.
So, in superconductor we need to follow this step carefully

Many-body berry curvature dipole

projector onto the many body ground state

$$\mathcal{D}^{\alpha\beta\gamma} = \frac{i\hbar^3}{e^3} \lim_{\mathbf{A} \rightarrow 0} \text{Tr}[P \partial_{\mathbf{A}}^{\alpha} (\partial_{\mathbf{A}}^{\beta} P) (\partial_{\mathbf{A}}^{\gamma} P)] - (\beta \leftrightarrow \gamma).$$

Capable of probing the many body dynamics of complex many body states

Oles Matsyshyn, Giovanni Vignale, **JS**, arXiv 2410.21363 (2024)

see also alternative description in terms of twisted boundary conditions from Resta

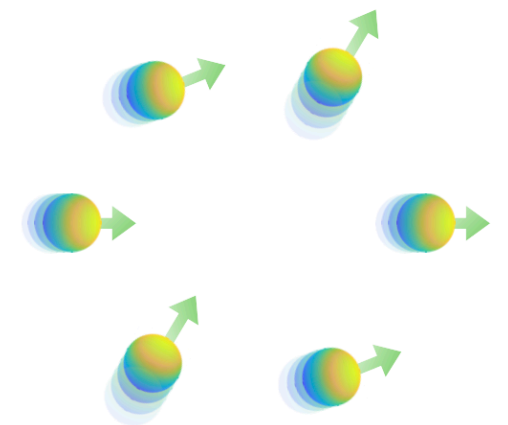
For a metal (using a Slater determinant Fermi sea)

$$\mathcal{D}_{\text{FS}}^{\alpha\beta\gamma} = \sum_{\{n, \mathbf{k}\} \in \text{FS}} i \partial_{\mathbf{k}}^{\alpha} \left[(\partial_{\mathbf{k}}^{\beta} \langle n, \mathbf{k} |) \partial_{\mathbf{k}}^{\gamma} | n, \mathbf{k} \rangle \right] - (\beta \leftrightarrow \gamma),$$

normal Bloch state (reproduces
Sodemann-Fu BCD)

$$\sum_{\mathbf{k}} [\Omega(\mathbf{k})]_d \partial_b f_0(\mathbf{k})$$

Sodemann and Fu PRL (2015)



Superconducting berry curvature dipole

projector onto the many body ground state

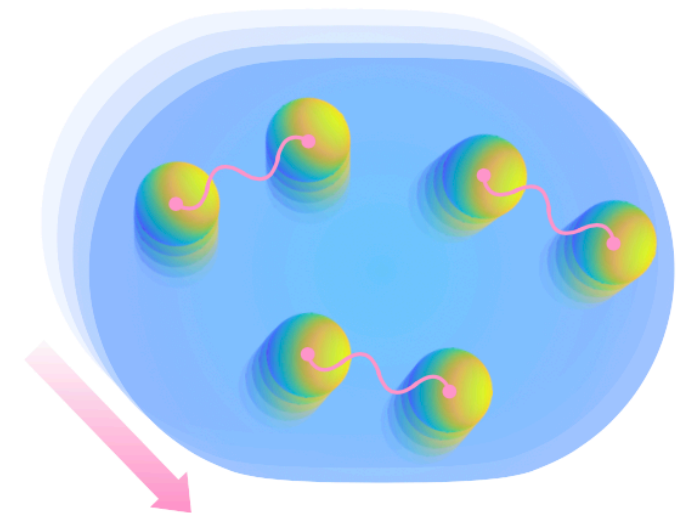
$$\mathcal{D}^{\alpha\beta\gamma} = \frac{i\hbar^3}{e^3} \lim_{\mathbf{A} \rightarrow 0} \text{Tr}[P \partial_{\mathbf{A}}^{\alpha} (\partial_{\mathbf{A}}^{\beta} P) (\partial_{\mathbf{A}}^{\gamma} P)] - (\beta \leftrightarrow \gamma).$$

COM Cooper pair momentum

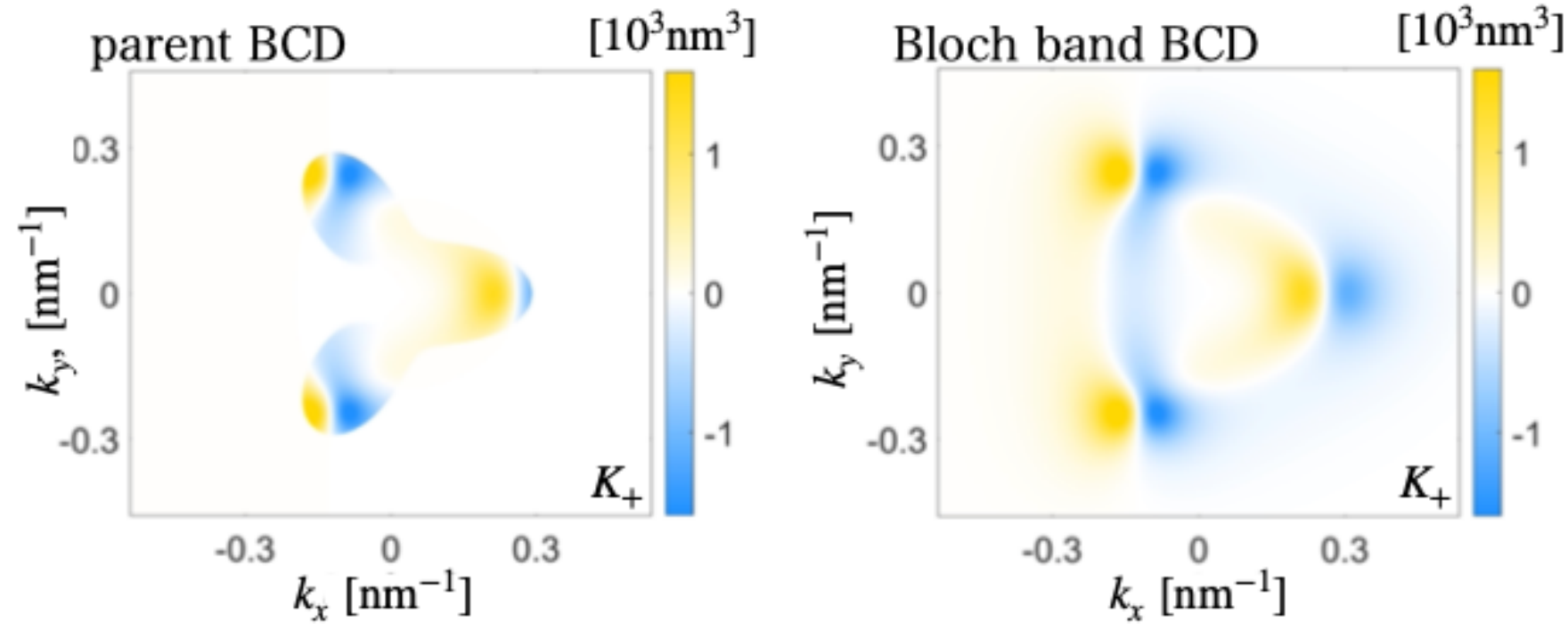
For a superconducting state: $|\text{SC}(\mathbf{A})\rangle = \prod_{\mathbf{k}} \left[u_{\mathbf{k},\mathbf{q}} + v_{\mathbf{k},\mathbf{q}} \hat{c}_{\mathbf{k}+\mathbf{q},\uparrow}^{\dagger}(\mathbf{A}) \hat{c}_{-\mathbf{k}+\mathbf{q},\downarrow}^{\dagger}(\mathbf{A}) \right] |0\rangle,$

$$\begin{aligned} \mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma} = & \int_{\mathbf{k}} \overbrace{\left[L_n^{\gamma}(\mathbf{k}, \mathbf{q}) \partial_{\mathbf{q}}^{\alpha} \partial_{\mathbf{q}}^{\beta} |v_{\mathbf{k},\mathbf{q}}|^2 - (\beta \leftrightarrow \gamma) \right]}^{\text{order parameter BCD distribution}}_{\mathbf{q} \rightarrow 0} \\ & + 2 \int_{\mathbf{k}} |v_{\mathbf{k},0}|^2 \underbrace{(\partial_{\mathbf{k}}^{\alpha} \partial_{\mathbf{k}}^{\beta} r_n^{\gamma}(\mathbf{k}) - \partial_{\mathbf{k}}^{\alpha} \partial_{\mathbf{k}}^{\gamma} r_n^{\beta}(\mathbf{k}))}_{\text{parent BCD distribution}}, \end{aligned}$$

Bloch state BCD distribution

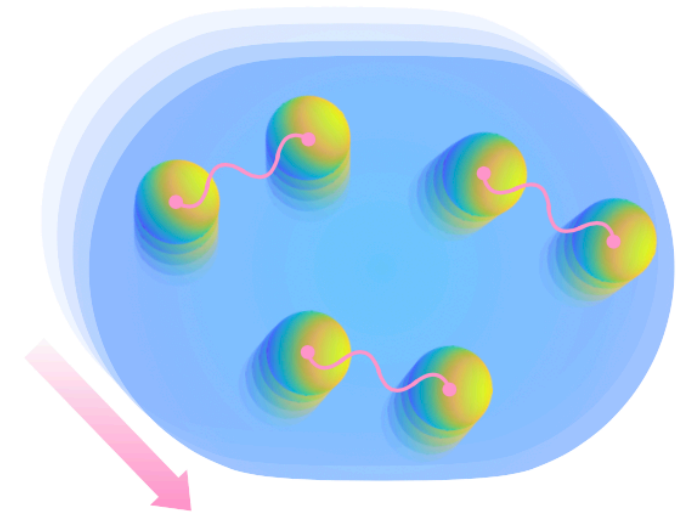


Parent BCD mimics the Bloch Band BCD



$$\begin{aligned}
 \mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma} = & \int_{\mathbf{k}} \overbrace{\left[L_n^\gamma(\mathbf{k}, \mathbf{q}) \partial_{\mathbf{q}}^\alpha \partial_{\mathbf{q}}^\beta |v_{\mathbf{k}, \mathbf{q}}|^2 - (\beta \leftrightarrow \gamma) \right]}^{\text{order parameter BCD distribution}}_{\mathbf{q} \rightarrow 0} \\
 & + 2 \int_{\mathbf{k}} |v_{\mathbf{k}, 0}|^2 \underbrace{(\partial_{\mathbf{k}}^\alpha \partial_{\mathbf{k}}^\beta r_n^\gamma(\mathbf{k}) - \partial_{\mathbf{k}}^\alpha \partial_{\mathbf{k}}^\gamma r_n^\beta(\mathbf{k}))}_{\text{parent BCD distribution}},
 \end{aligned}$$

Bloch state BCD distribution



Order parameter BCD

projector onto the many body ground state

$$\mathcal{D}^{\alpha\beta\gamma} = \frac{i\hbar^3}{e^3} \lim_{\mathbf{A} \rightarrow 0} \text{Tr}[P \partial_{\mathbf{A}}^{\alpha} (\partial_{\mathbf{A}}^{\beta} P) (\partial_{\mathbf{A}}^{\gamma} P)] - (\beta \leftrightarrow \gamma).$$

Berry connection of the Bloch electron at $\mathbf{k}+\mathbf{q}$ and $-\mathbf{k}+\mathbf{q}$

Depends on structure of the pairing:

$$\mathbf{L}_n(\mathbf{k}, \mathbf{q}) = -\partial_{\mathbf{q}}^{\gamma} \arg[\langle n, \mathbf{k} + \mathbf{q} | \hat{\Delta} | n, -\mathbf{k} + \mathbf{q} \rangle^*] + r_n^{\gamma}(-\mathbf{k} + \mathbf{q}) + r_n^{\gamma}(\mathbf{k} + \mathbf{q})$$

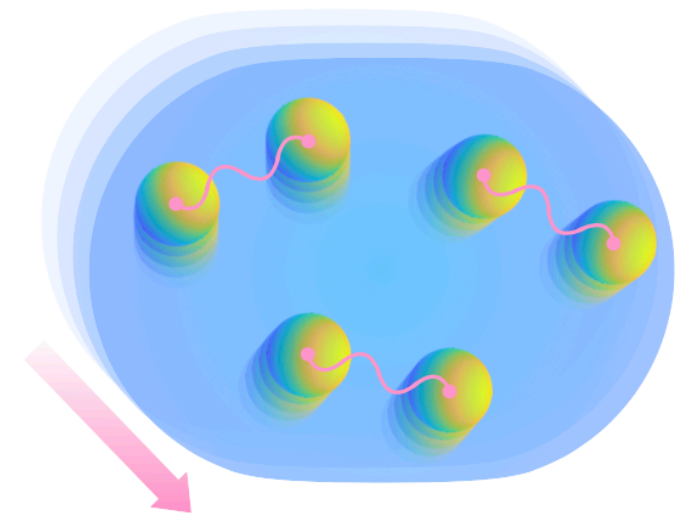
order parameter BCD distribution

$$\mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma} = \int_{\mathbf{k}} \left[L_n^{\gamma}(\mathbf{k}, \mathbf{q}) \partial_{\mathbf{q}}^{\alpha} \partial_{\mathbf{q}}^{\beta} |v_{\mathbf{k}, \mathbf{q}}|^2 - (\beta \leftrightarrow \gamma) \right]_{\mathbf{q} \rightarrow 0}$$

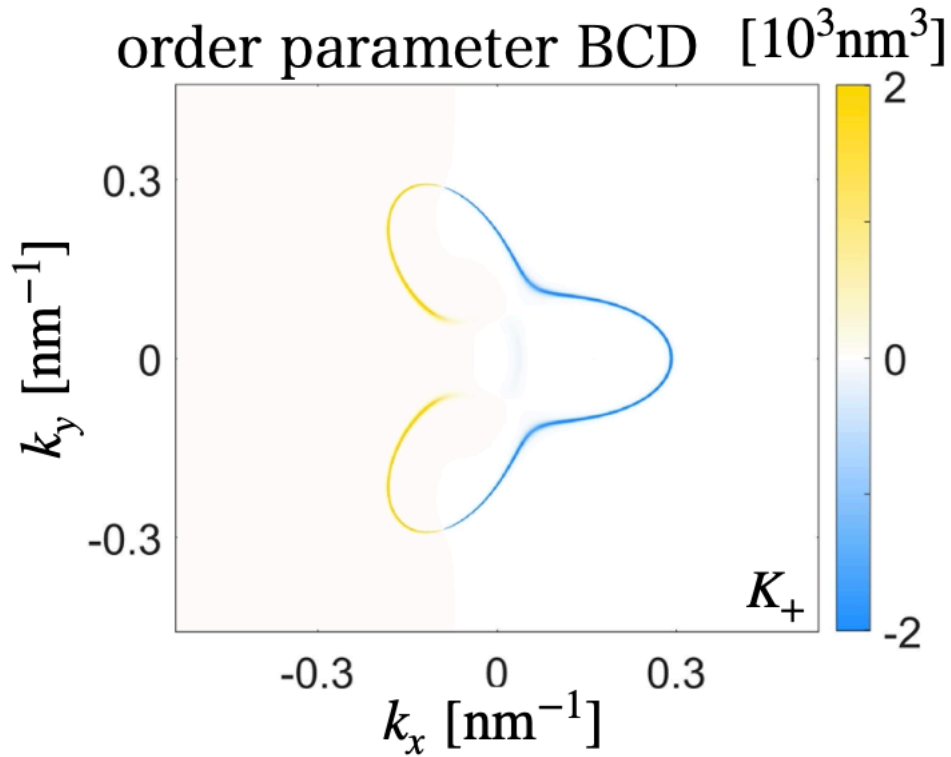
Bloch state BCD distribution

$$+ 2 \int_{\mathbf{k}} |v_{\mathbf{k}, 0}|^2 \left(\partial_{\mathbf{k}}^{\alpha} \partial_{\mathbf{k}}^{\beta} r_n^{\gamma}(\mathbf{k}) - \partial_{\mathbf{k}}^{\alpha} \partial_{\mathbf{k}}^{\gamma} r_n^{\beta}(\mathbf{k}) \right),$$

parent BCD distribution

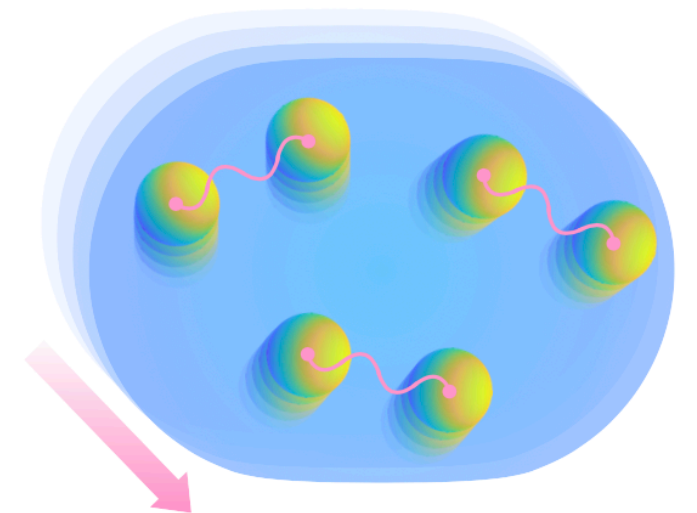


Order parameter BCD is **emergent**



$$\begin{aligned}
 \mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma} = & \int_{\mathbf{k}} \overbrace{\left[L_n^\gamma(\mathbf{k}, \mathbf{q}) \partial_{\mathbf{q}}^\alpha \partial_{\mathbf{q}}^\beta |v_{\mathbf{k}, \mathbf{q}}|^2 - (\beta \leftrightarrow \gamma) \right]}^{\text{order parameter BCD distribution}}_{\mathbf{q} \rightarrow 0} \\
 & + 2 \int_{\mathbf{k}} \underbrace{|v_{\mathbf{k}, 0}|^2 (\partial_{\mathbf{k}}^\alpha \partial_{\mathbf{k}}^\beta r_n^\gamma(\mathbf{k}) - \partial_{\mathbf{k}}^\alpha \partial_{\mathbf{k}}^\gamma r_n^\beta(\mathbf{k}))}_{\text{parent BCD distribution}},
 \end{aligned}$$

Bloch state BCD distribution



Cooper pair shift

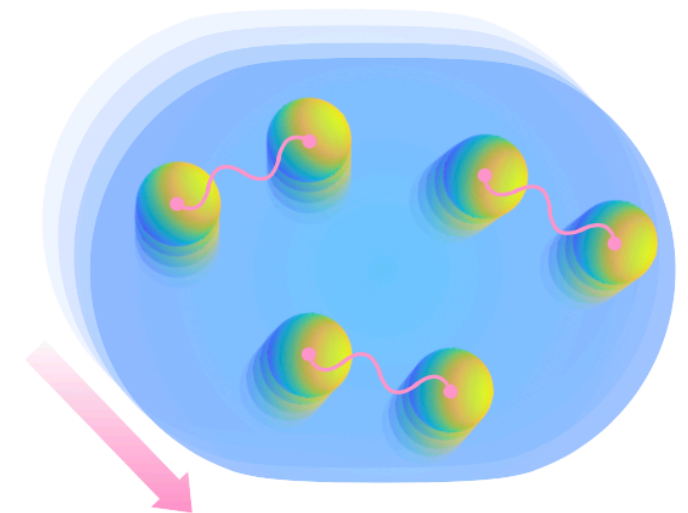
Depends on structure of the pairing:

$$\mathbf{L}_n(\mathbf{k}, \mathbf{q}) = -\partial_{\mathbf{q}}^\gamma \arg[\langle n, \mathbf{k} + \mathbf{q} | \hat{\Delta} | n, -\mathbf{k} + \mathbf{q} \rangle^*] + r_n^\gamma(-\mathbf{k} + \mathbf{q}) + r_n^\gamma(\mathbf{k} + \mathbf{q})$$

Berry connection of the
Bloch electron at $\mathbf{k}+\mathbf{q}$ and
 $-\mathbf{k}+\mathbf{q}$



By pairing, there is spatial position of the center of mass shifts



Cooper pair shift

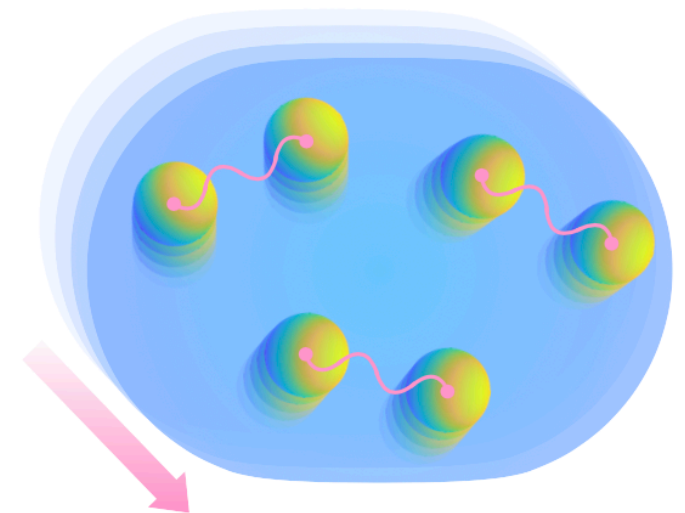
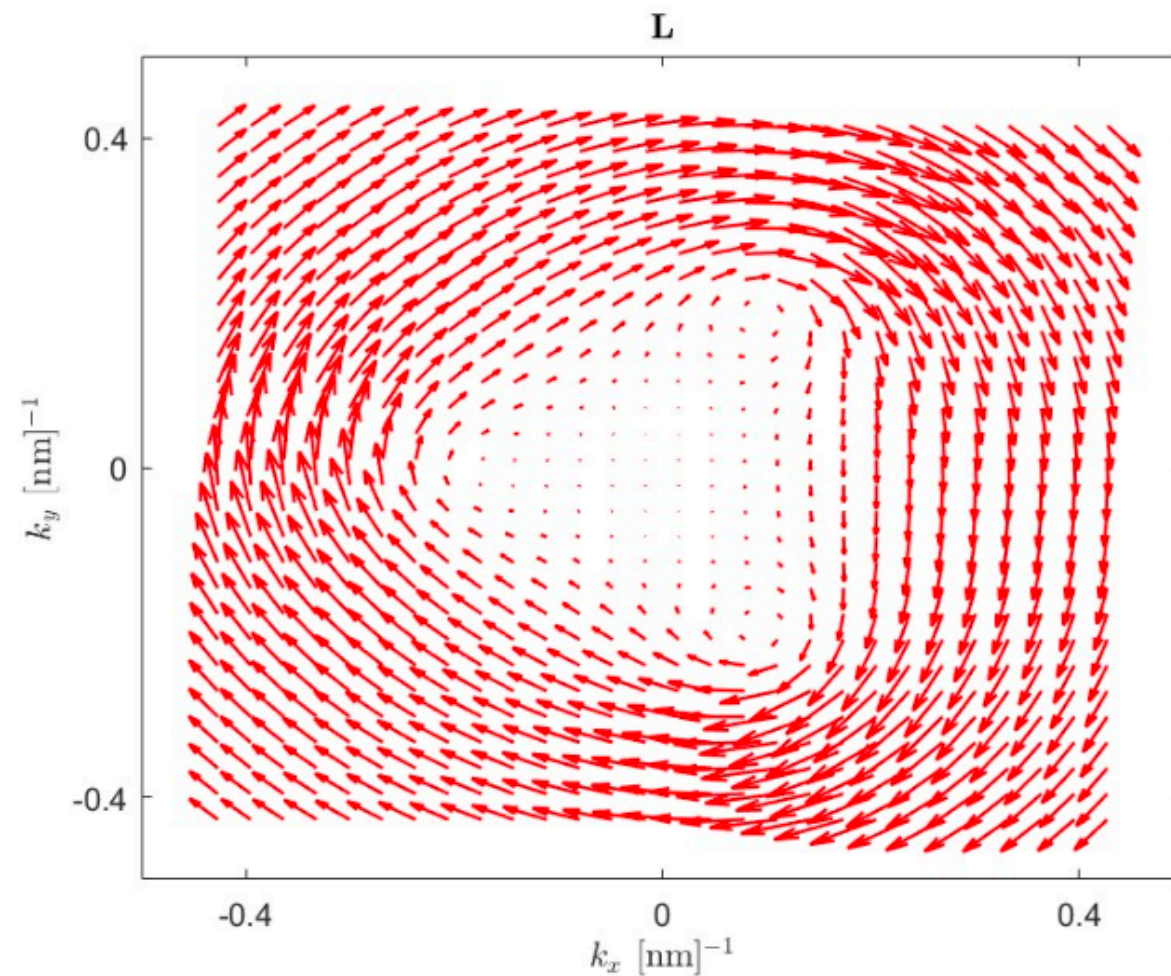
Depends on structure of the pairing:

$$\mathbf{L}_n(\mathbf{k}, \mathbf{q}) = -\partial_{\mathbf{q}}^\gamma \arg[\langle n, \mathbf{k} + \mathbf{q} | \hat{\Delta} | n, -\mathbf{k} + \mathbf{q} \rangle^*] + r_n^\gamma(-\mathbf{k} + \mathbf{q}) + r_n^\gamma(\mathbf{k} + \mathbf{q})$$

Berry connection of the
Bloch electron at $\mathbf{k}+\mathbf{q}$ and
 $-\mathbf{k}+\mathbf{q}$



By pairing, there is spatial position of the center of mass shifts



Cooper pair shift

Depends on structure of the pairing:

$$\mathbf{L}_n(\mathbf{k}, \mathbf{q}) = -\partial_{\mathbf{q}}^{\gamma} \arg[\langle n, \mathbf{k} + \mathbf{q} | \hat{\Delta} | n, -\mathbf{k} + \mathbf{q} \rangle^*] + r_n^{\gamma}(-\mathbf{k} + \mathbf{q}) + r_n^{\gamma}(\mathbf{k} + \mathbf{q})$$

Berry connection of the
Bloch electron at $\mathbf{k}+\mathbf{q}$ and
 $-\mathbf{k}+\mathbf{q}$



By pairing, there is spatial position of the center of mass shifts

In TRS symmetric system and at $\mathbf{q}=0$, simple relationship for $\mathbf{L}$:

$$\mathbf{L}_n(\mathbf{k}, \mathbf{q}) = -\partial_{\mathbf{q}} \arg \left[\frac{\langle n, \mathbf{k} + \mathbf{q} | \hat{\Delta} | n, \mathbf{k} - \mathbf{q} \rangle}{\langle n, \mathbf{k} + \mathbf{q} | n, \mathbf{k} - \mathbf{q} \rangle} \right].$$

no orbital structure

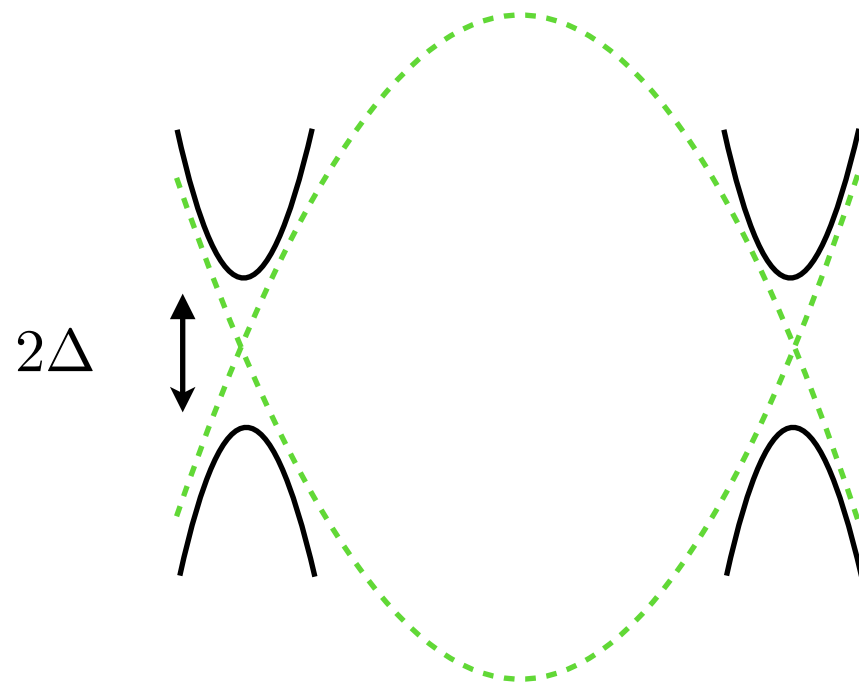
$$\hat{\Delta} = \mathbf{1} \quad \mathbf{L} = 0$$

with orbital structure

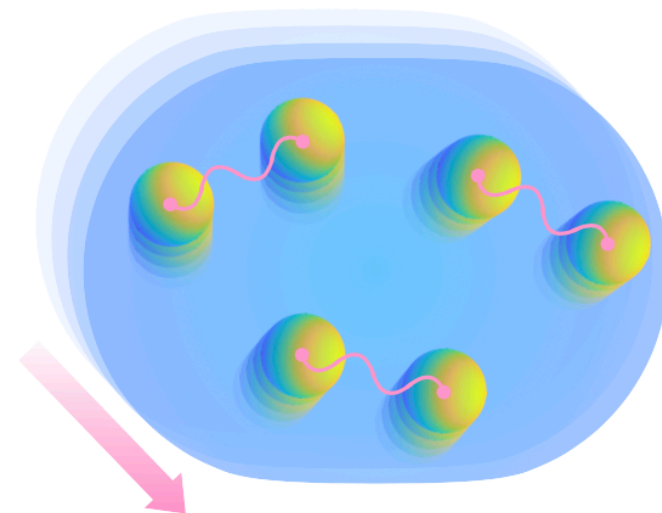
$$\mathbf{L} \neq 0$$

Dissipationless superconductor BCD responses

Focus on frequencies $\omega \ll \Delta$



$\omega \ll \Delta$



for such low frequencies, obtain non-dissipative dynamics: response of coherent many body state!

Adiabatic dynamics under the gap

velocity - momentum relationship

$$\mathbf{v} = (1/m)\mathbf{p}$$

oscillating electric fields can modulate the collective
“effective inverse mass” of Cooper pairs

collective velocity

$$\delta v^\gamma(t) = \frac{\partial^2 \mathcal{E}(\mathbf{q})}{\partial q^\alpha \partial q^\gamma} \frac{\delta q^\gamma(t)}{V\hbar} + \delta \dot{q}^\alpha(t) \mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma} \delta q^\beta(t),$$

superfluid weight

composed of inverse mass of electrons as well as “quantum metric”
contribution (Nat Comms 9944 (2015))

superconducting Berry curvature dipole

Adiabatic dynamics under the gap

velocity

$$\mathbf{v}(\mathbf{q}) = \frac{1}{\hbar} \int_{\mathbf{k}} \sum_a f_a (\partial_{\mathbf{q}} \varepsilon_{\mathbf{k},a}(\mathbf{q}) - \hbar \dot{\mathbf{q}} \times [\Omega_{\mathbf{k}}(\mathbf{q})])$$

oscillating electric fields can modulate the collective
“effective inverse mass” of Cooper pairs

collective velocity

$$\delta v^\gamma(t) = \frac{\partial^2 \mathcal{E}(\mathbf{q})}{\partial q^\alpha \partial q^\gamma} \frac{\delta q^\gamma(t)}{V \hbar} + \delta \dot{q}^\alpha(t) \mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma} \delta q^\beta(t),$$

superfluid weight

composed of inverse mass of electrons as well as “quantum metric”
contribution (Nat Comms 9944 (2015))

superconducting Berry curvature dipole

Adiabatic dynamics under the gap

“extra inertia due to pairing”

$$\mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma} = \underbrace{\int_{\mathbf{k}} \left[L_n^\gamma(\mathbf{k}, \mathbf{q}) \partial_{\mathbf{q}}^\alpha \partial_{\mathbf{q}}^\beta |v_{\mathbf{k}, \mathbf{q}}|^2 - (\beta \leftrightarrow \gamma) \right]_{\mathbf{q} \rightarrow 0}}_{\text{order parameter BCD distribution}} + 2 \int_{\mathbf{k}} \underbrace{|v_{\mathbf{k}, 0}|^2 (\partial_{\mathbf{k}}^\alpha \partial_{\mathbf{k}}^\beta r_n^\gamma(\mathbf{k}) - \partial_{\mathbf{k}}^\alpha \partial_{\mathbf{k}}^\gamma r_n^\beta(\mathbf{k}))}_{\text{parent BCD distribution}},$$

Bloch state BCD distribution

collective velocity

$$\delta v^\gamma(t) = \frac{\partial^2 \mathcal{E}(\mathbf{q})}{\partial q^\alpha \partial q^\gamma} \frac{\delta q^\gamma(t)}{V \hbar} + \delta \dot{q}^\alpha(t) \mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma} \delta q^\beta(t),$$

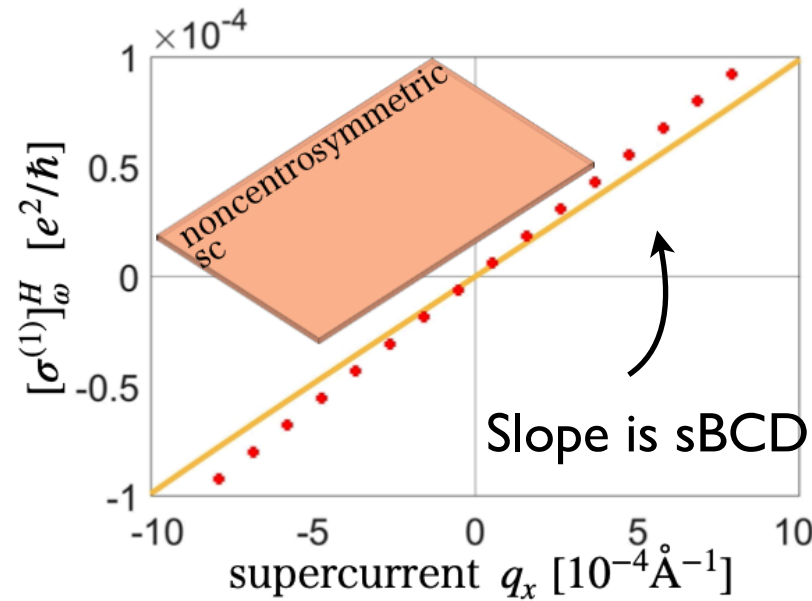
superfluid weight

composed of inverse mass of electrons as well as “quantum metric”
contribution (Nat Comms 9944 (2015))

superconducting Berry curvature dipole

Superconductor BCD mediated responses

Noncentrosymmetric
Superconductor



$$j_t^\gamma = j_0^\gamma + \int_{\omega} e^{-i\omega t} \left\{ [\sigma^{(1)}]_{\omega}^{\gamma\beta} E_{\omega}^{\beta} + \int_{\omega_1} [\sigma^{(2)}]_{\omega, \omega_1}^{\gamma\alpha\beta} E_{\omega_1}^{\alpha} E_{\omega - \omega_1}^{\beta} \right\},$$

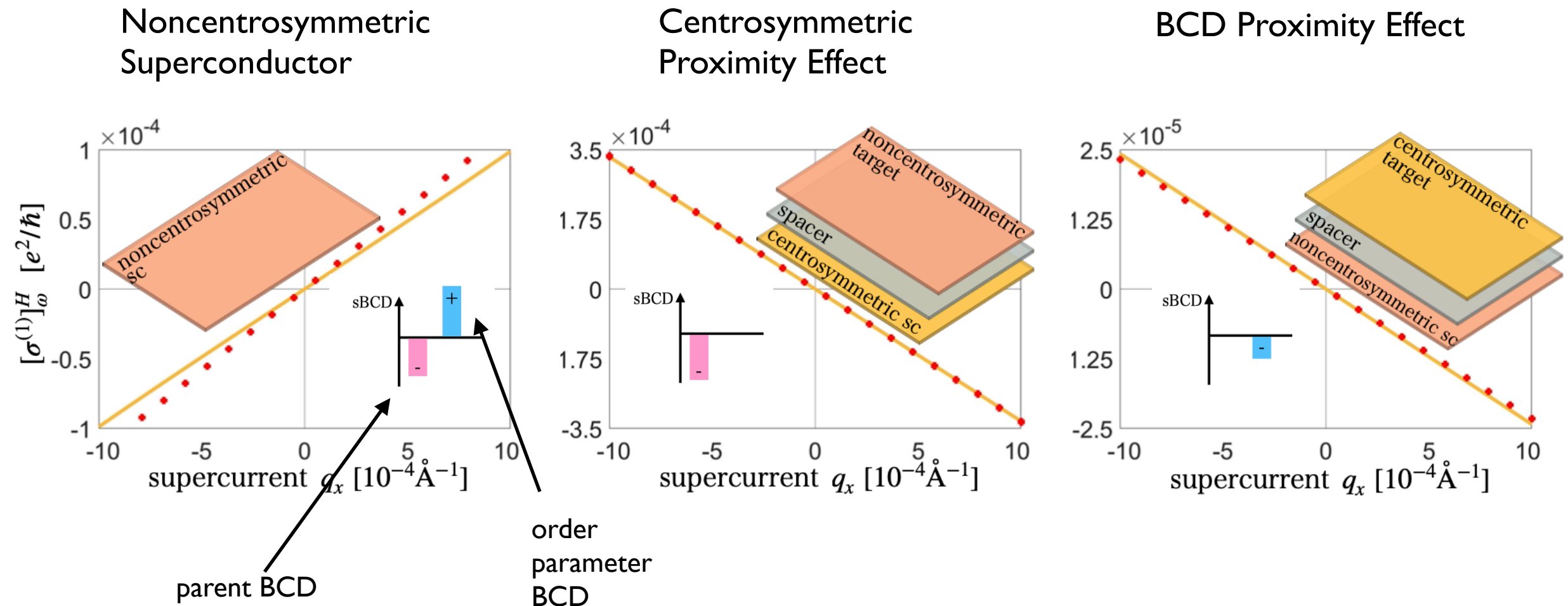
AC Hall conductivity: generates a Kerr rotation effect

$$[\sigma^{(1)}]_{\omega}^H = e[\hat{\rho}^{-1} \mathbf{j}_0]^{\alpha} \mathcal{D}_{\text{sBCD}}^{\alpha xy},$$

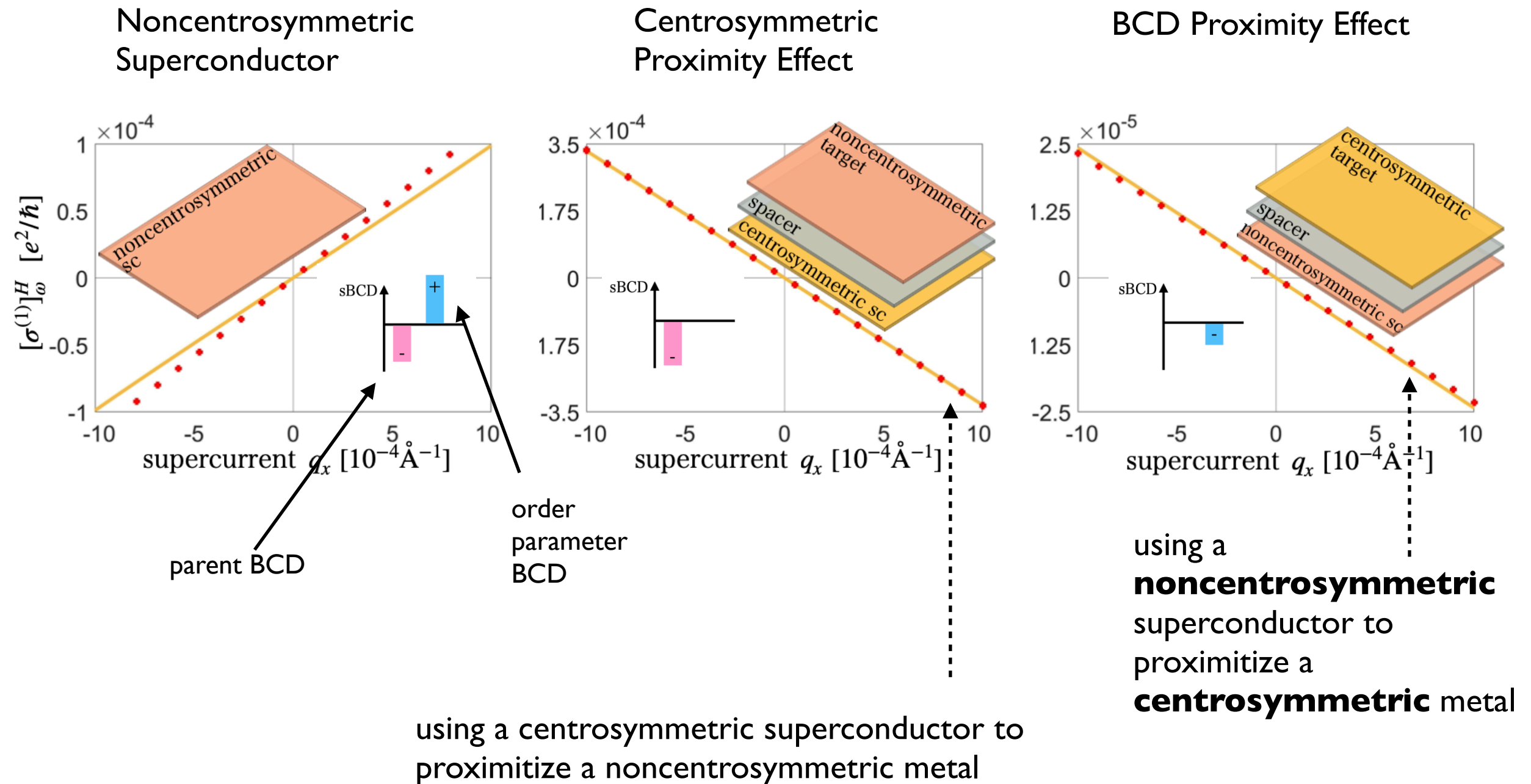
Oles Matsyshyn, Giovanni Vignale, **JS**, arXiv 2410.21363 (2024)

* Here we have used gated rhombohedral trilayer graphene (RTG) that enters into a superconducting state. Yellow is sBCD-bogoliubov formula shown previously. Dots are full multiband theory

Superconductor BCD mediated AC Hall conductivity

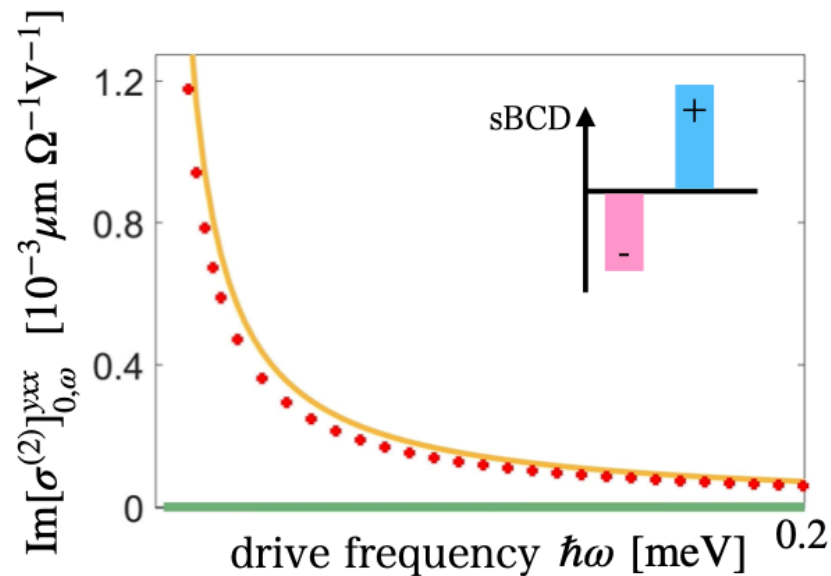
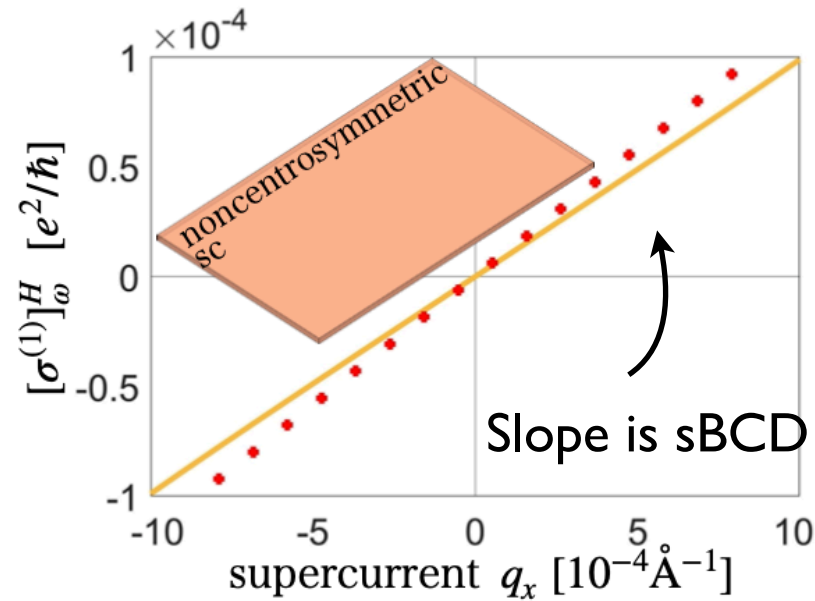


Superconductor BCD mediated AC Hall conductivity



Superconductor BCD mediated responses

Noncentrosymmetric
Superconductor



$$j_t^\gamma = j_0^\gamma + \int_{\omega} e^{-i\omega t} \left\{ [\sigma^{(1)}]_{\omega}^{\gamma\beta} E_{\omega}^{\beta} + \int_{\omega_1} [\sigma^{(2)}]_{\omega, \omega_1}^{\gamma\alpha\beta} E_{\omega_1}^{\alpha} E_{\omega - \omega_1}^{\beta} \right\},$$

AC Hall conductivity: generates a Kerr rotation effect

$$[\sigma^{(1)}]_{\omega}^H = e[\hat{\rho}^{-1} \mathbf{j}_0]^{\alpha} \mathcal{D}_{\text{sBCD}}^{\alpha xy},$$

second order nonlinear photocurrent

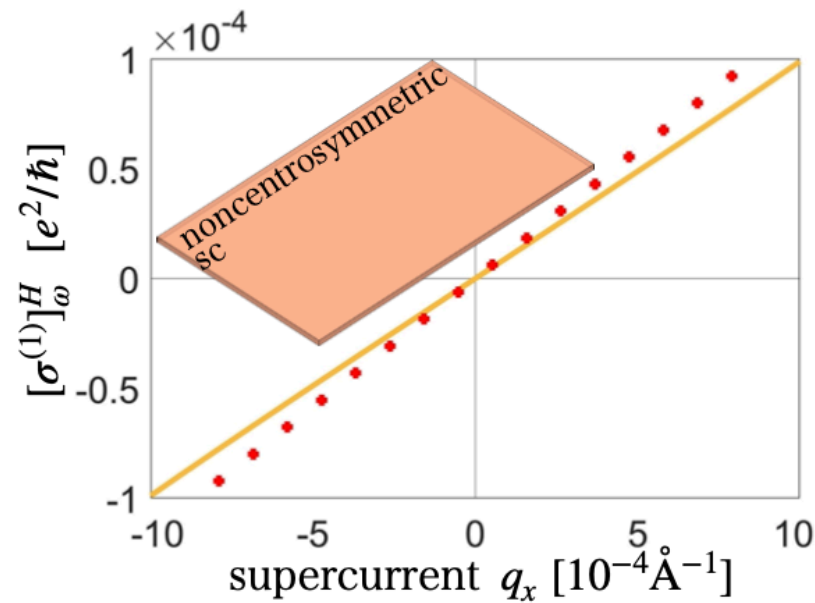
$$[\sigma^{(2)}]_{\omega, \omega_1}^{\gamma\alpha\beta} = \frac{ie^3}{\hbar^2} \frac{\mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma}}{\omega_1},$$

Oles Matsyshyn, Giovanni Vignale, **JS**, arXiv 2410.21363 (2024)

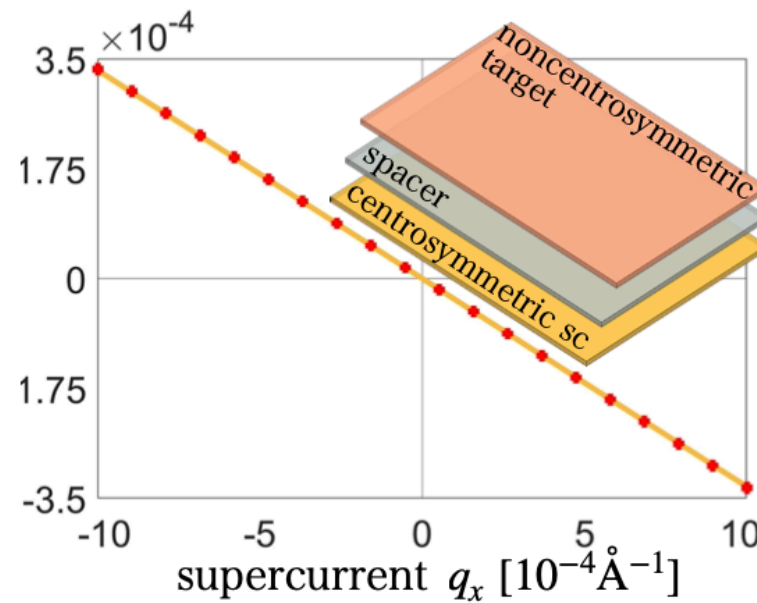
* Here we have used gated rhombohedral trilayer graphene (RTG) that enters into a superconducting state. Yellow is sBCD-bogoliubov formula shown previously. Dots are full multiband theory

Superconductor BCD responses

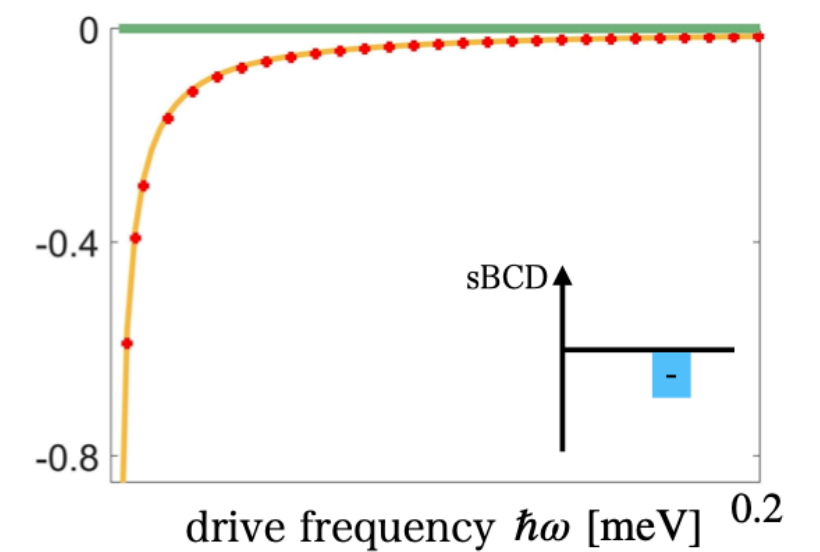
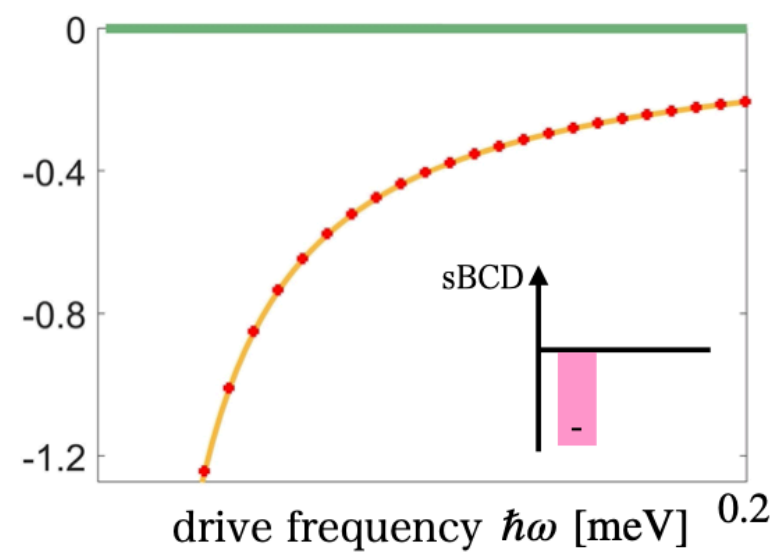
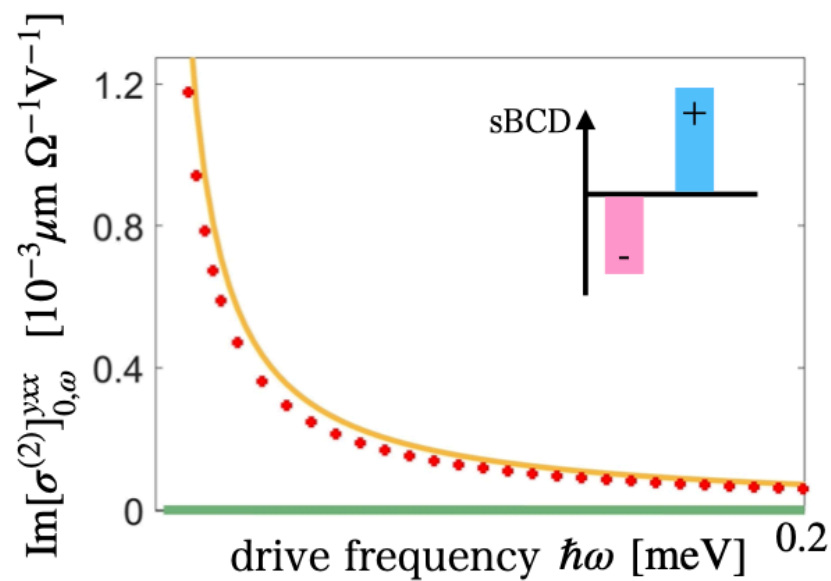
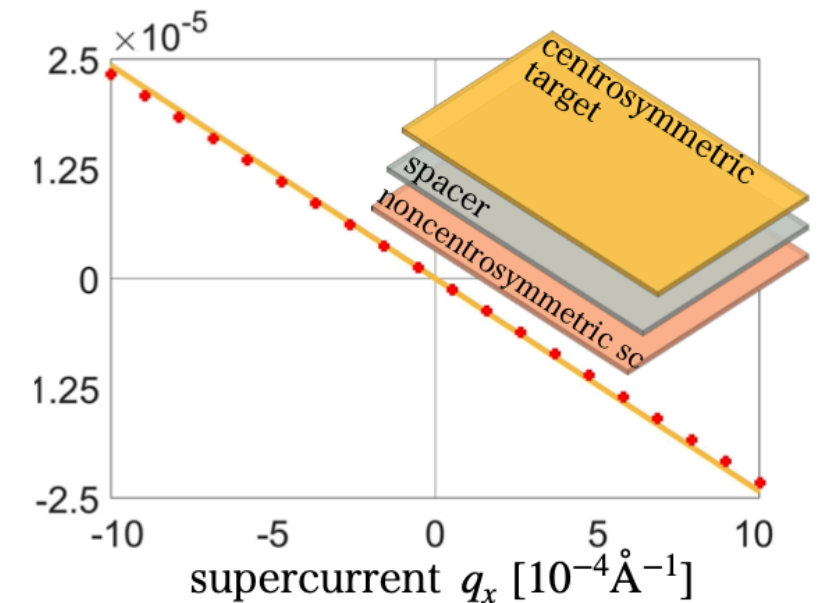
Noncentrosymmetric
Superconductor



Centrosymmetric
Proximity Effect



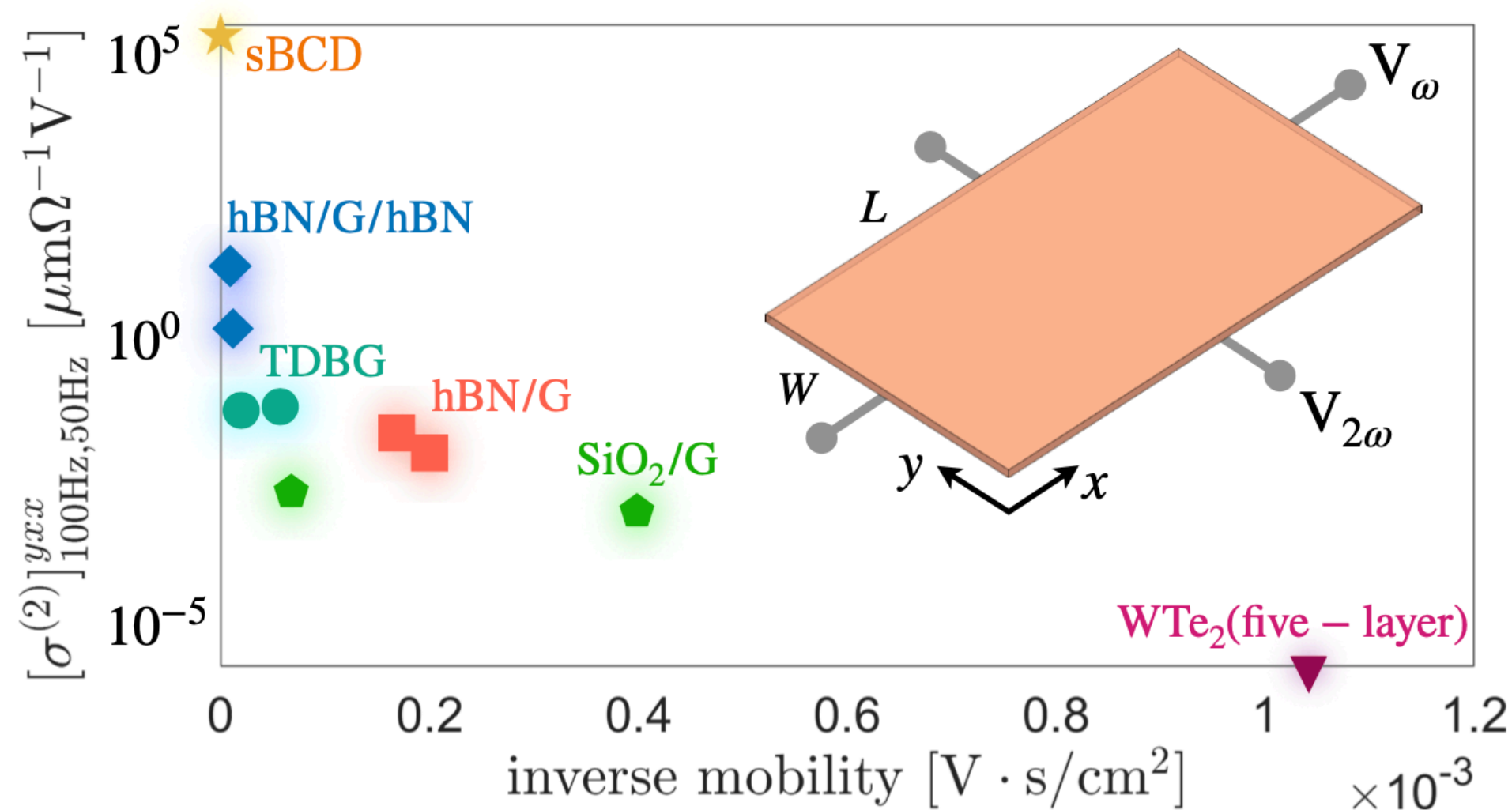
BCD Proximity Effect



undergap rectified second order nonlinearities in superconductors were also recently analyzed by Watanabe & Yanase PRB 107, 024513 (2022)

Superconductor BCD nonlinear Hall effect

Frequency insensitive nonlinear Hall effect: signal depends on sBCD and superfluid weight

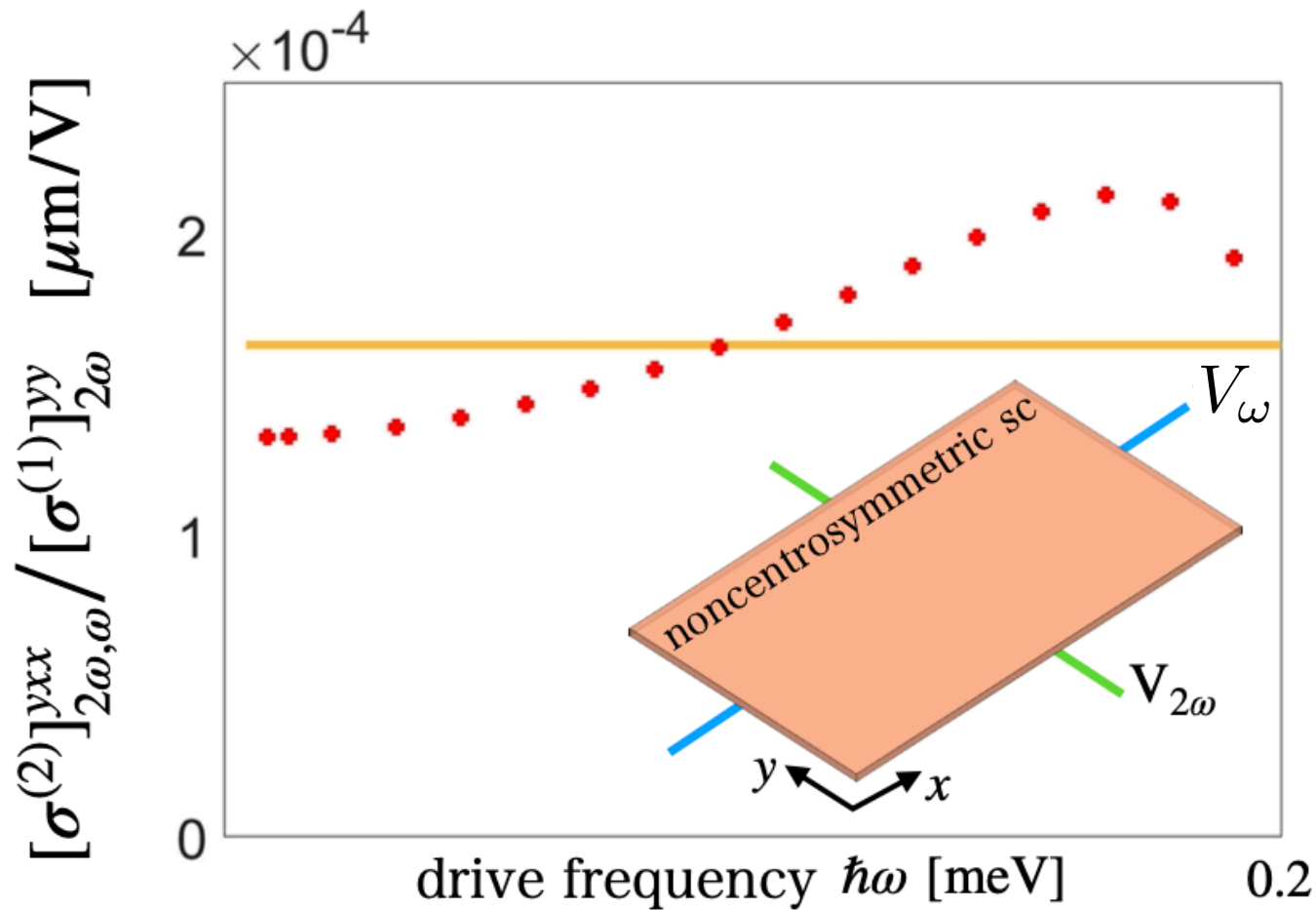


Superconductor BCD nonlinear Hall effect

Frequency insensitive nonlinear Hall effect: signal depends on sBCD and superfluid weight

controlled by
superfluid weight

controlled by sBCD



Impact of symmetries: more generally

longitudinal response (linear polarization)

chiral response (circular polarization)

$$j^a = \text{Re}\chi_{abc} \text{Re} \left(\overset{\downarrow}{E_\omega^b} (E_\omega^c)^* \right) - \text{Im}\chi_{abc} \text{Im} \left(E_\omega^b \overset{\downarrow}{(E_\omega^c)^*} \right).$$

Time-reversal operation

$$-j^a = \text{Re}\chi_{abc} \text{Re} \left(E_\omega^b (E_\omega^c)^* \right) + \text{Im}\chi_{abc} \text{Im} \left(E_\omega^b (E_\omega^c)^* \right).$$

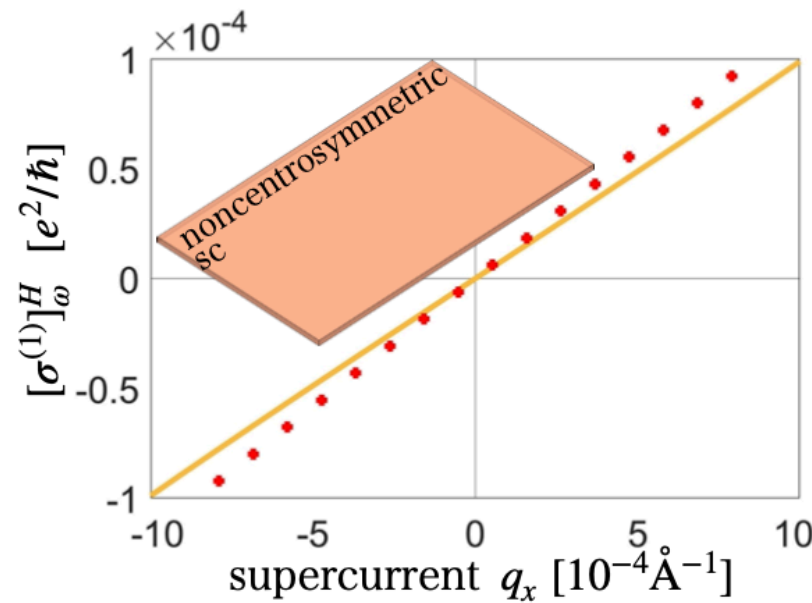
In the presence of time-reversal symmetry (without dissipation)

$\text{Re}\chi_{abc}$ vanishes

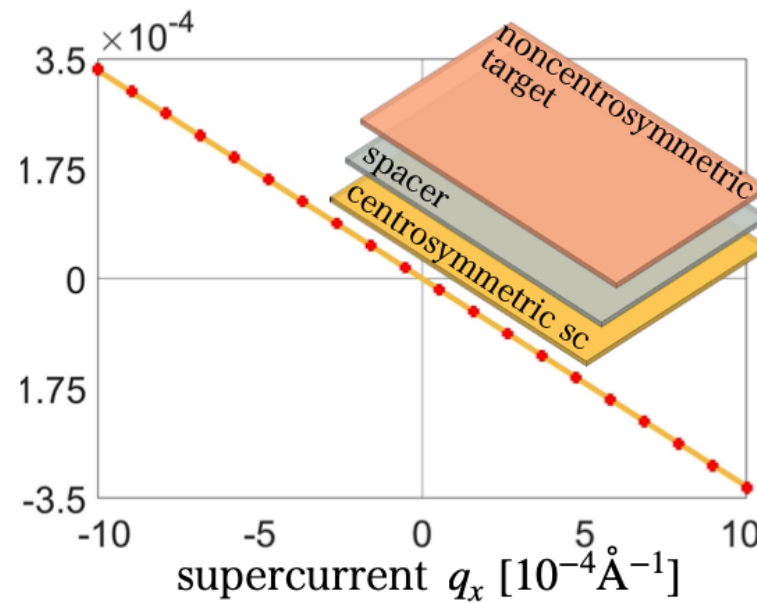
$\text{Im}\chi_{abc}$ allowed

Dissipationless photocurrent in a superconductor

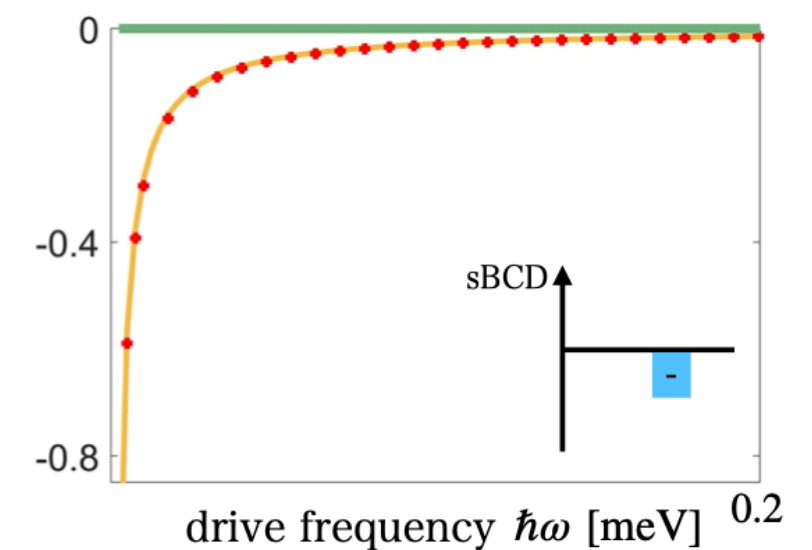
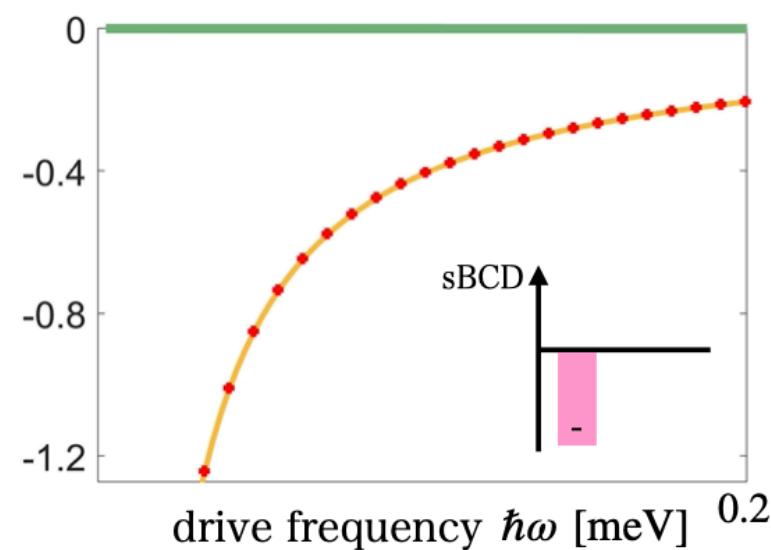
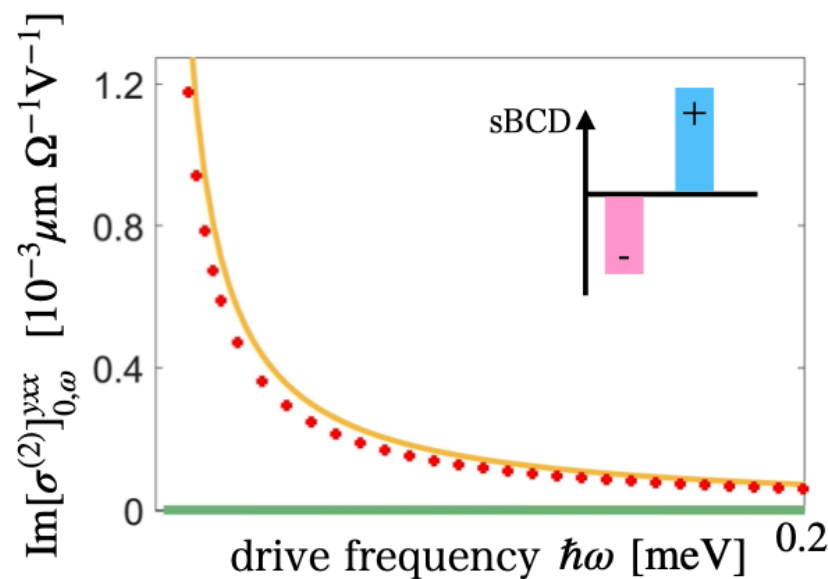
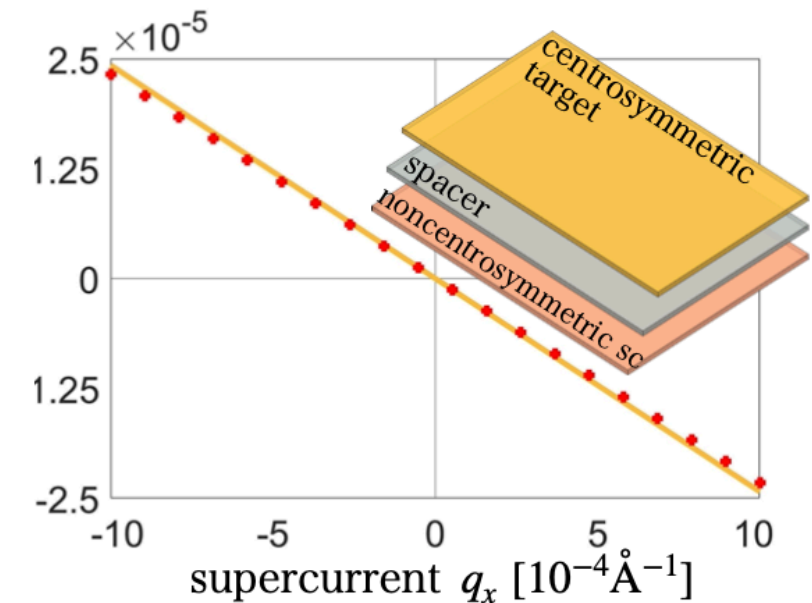
Noncentrosymmetric
Superconductor



Centrosymmetric
Proximity Effect

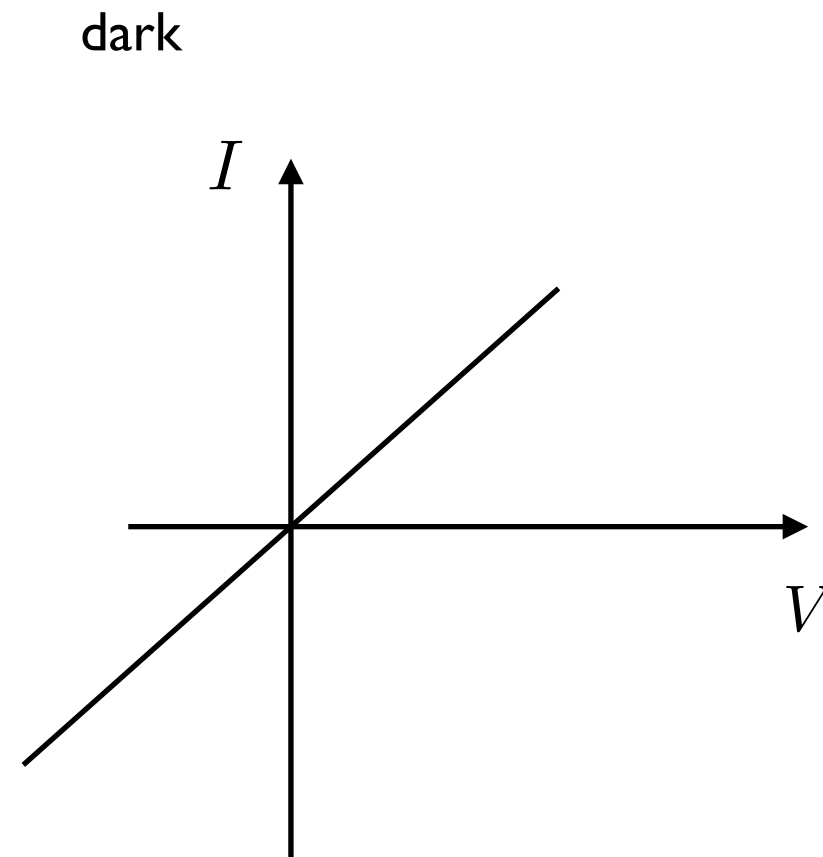
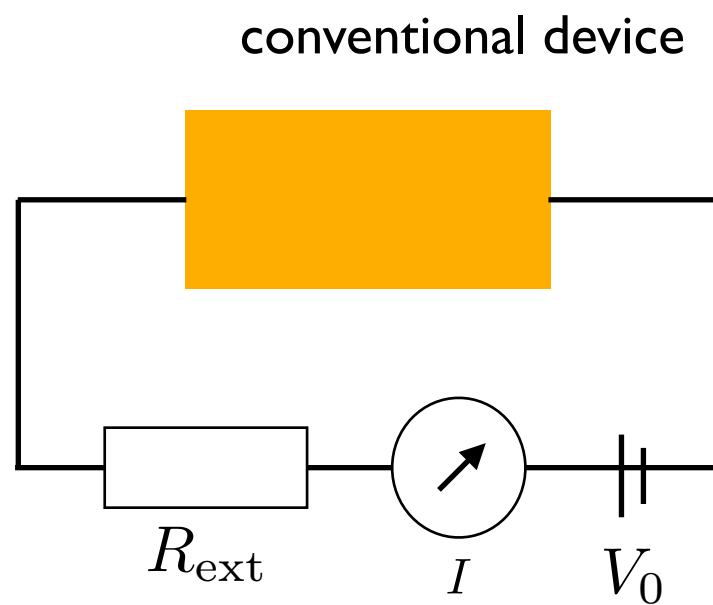


BCD Proximity Effect

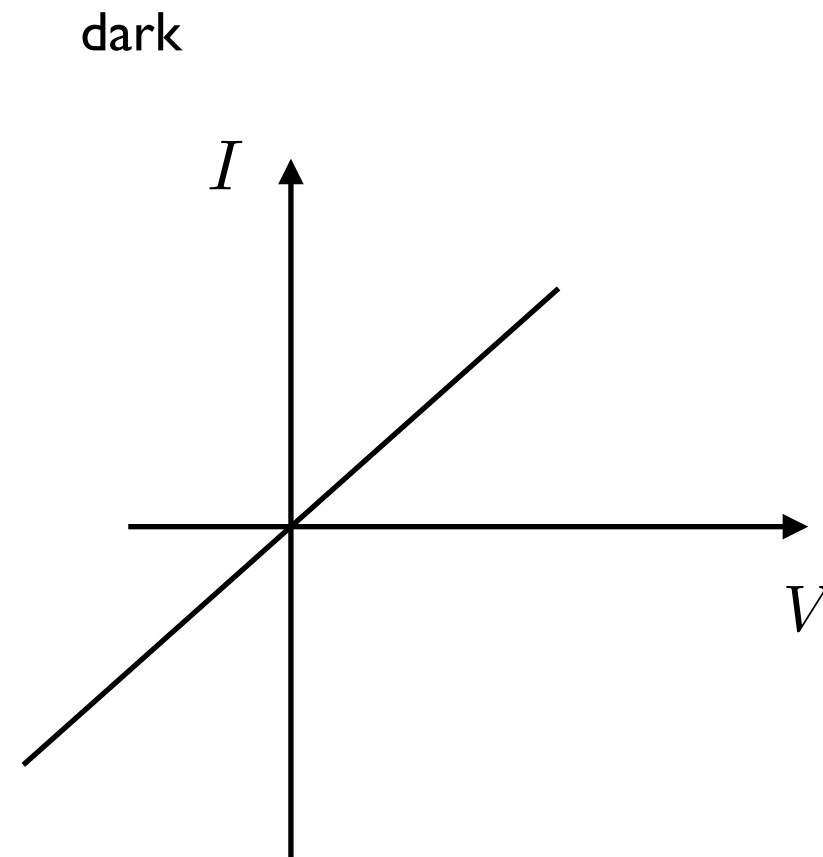
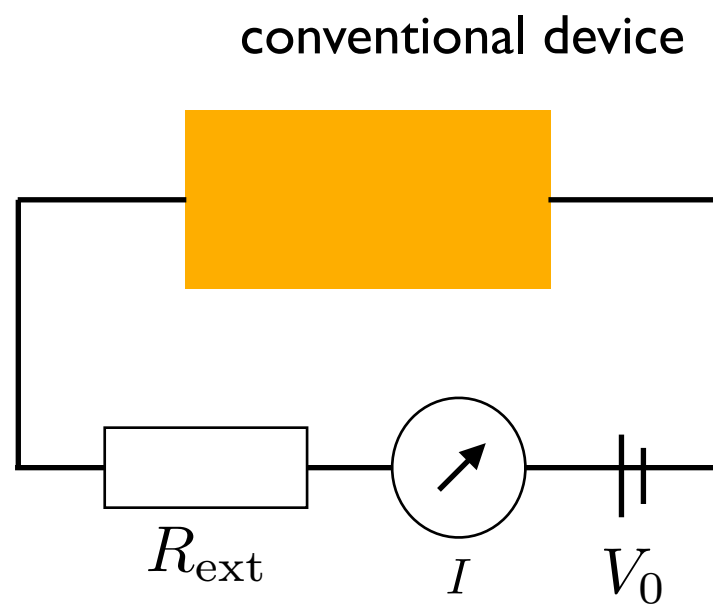


undergap rectified second order nonlinearities in superconductors were also recently analyzed by Watanabe & Yanase PRB 107, 024513 (2022)

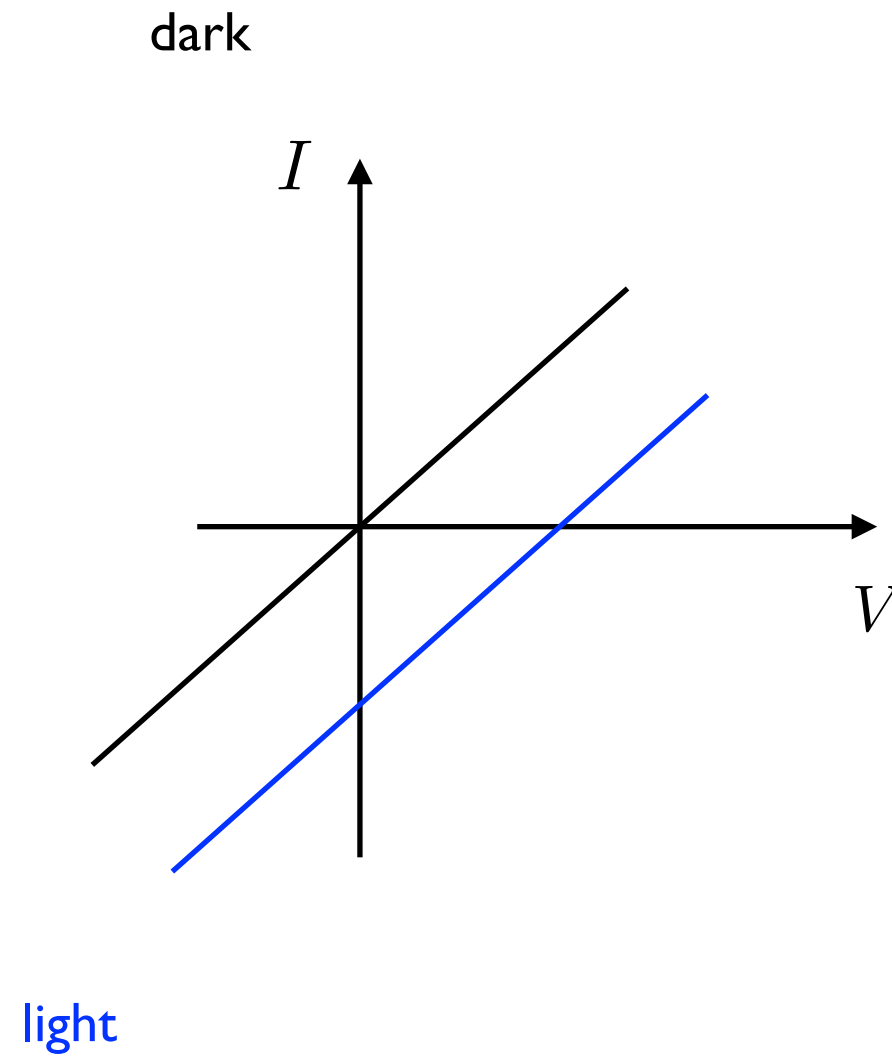
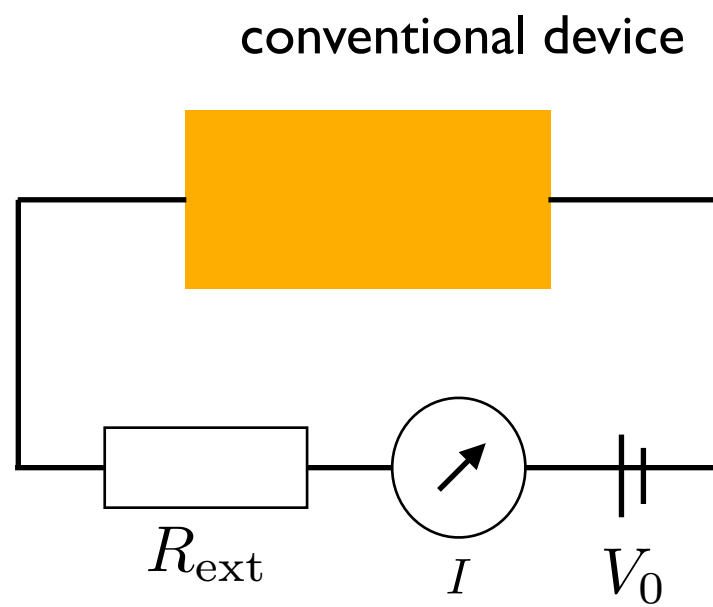
Bonus: How to measure a photocurrent in a superconducting device?



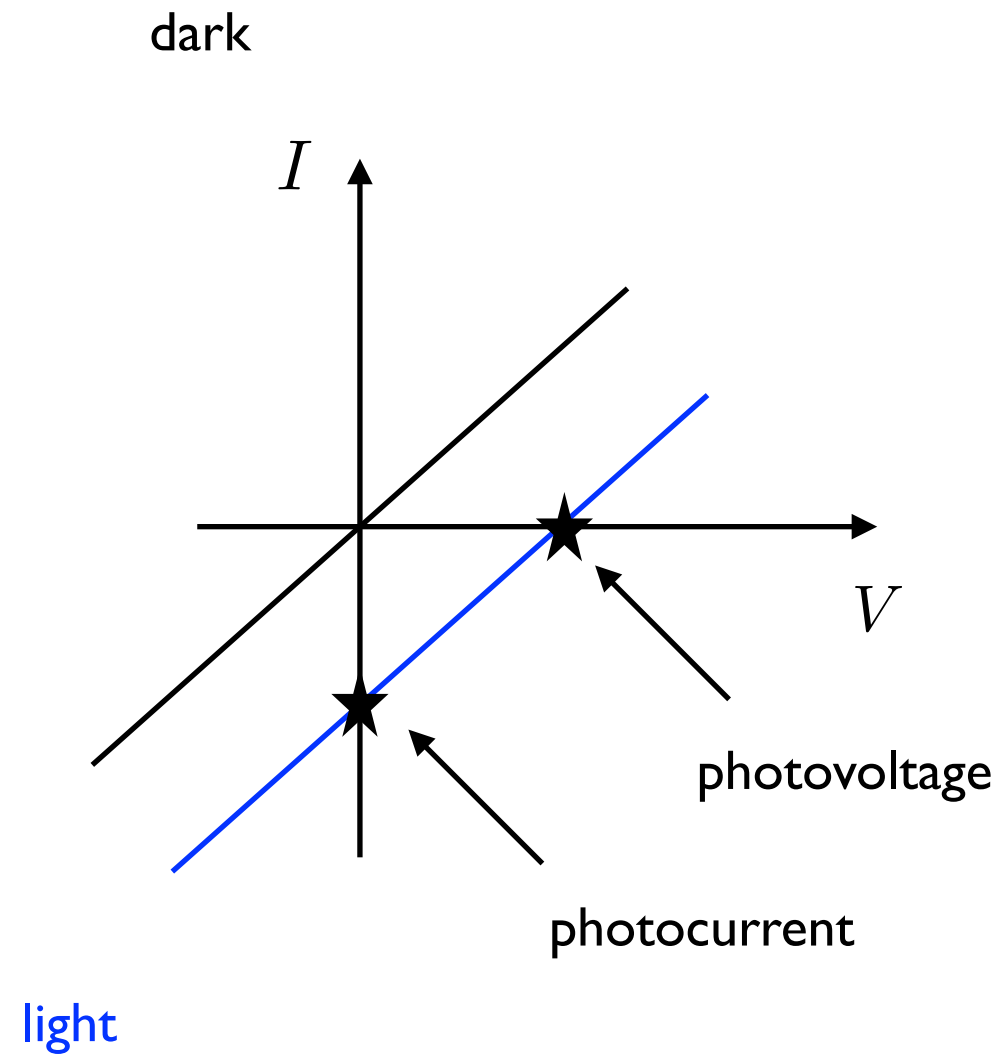
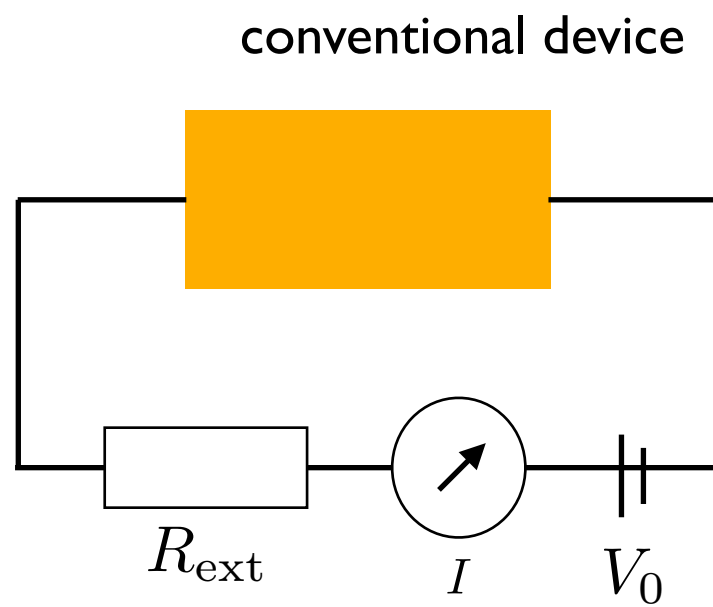
How to measure a photocurrent in a device



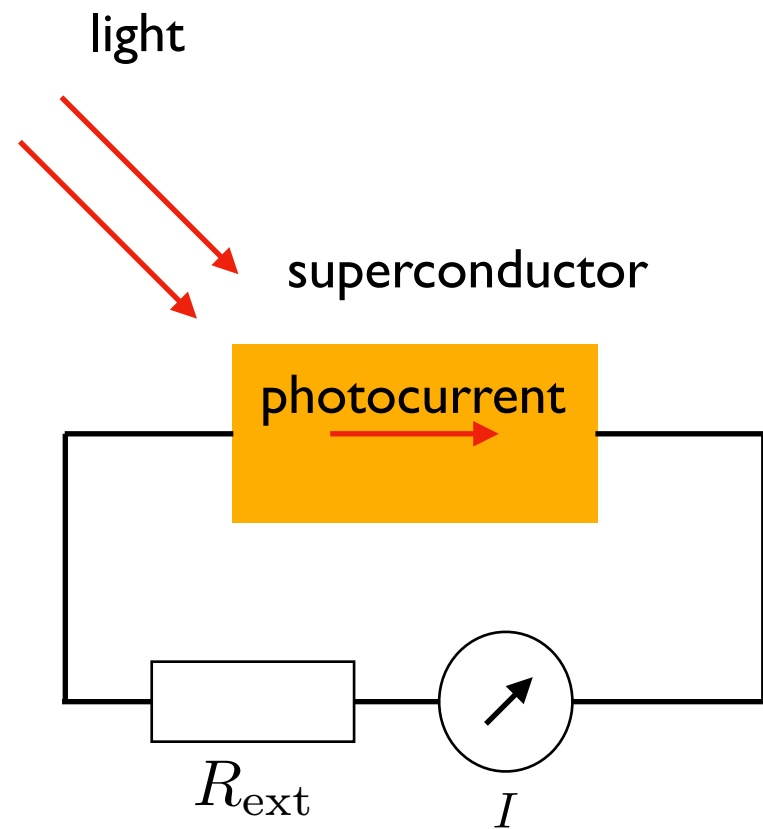
How to measure a photocurrent in a device



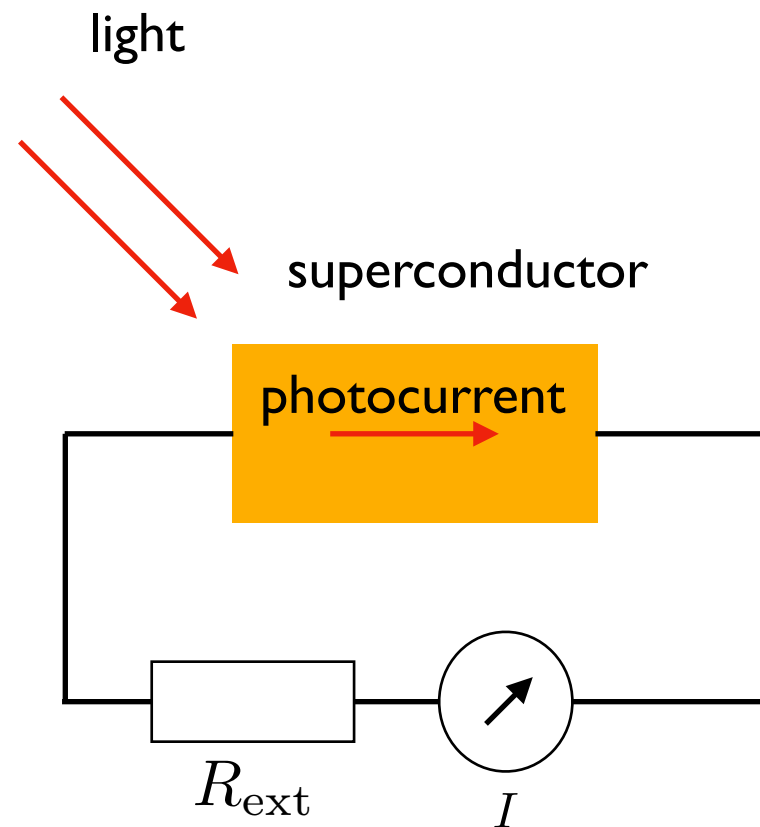
How to measure a photocurrent in a device



How to measure a photocurrent in a superconductor?



How to measure a photocurrent in a superconductor?



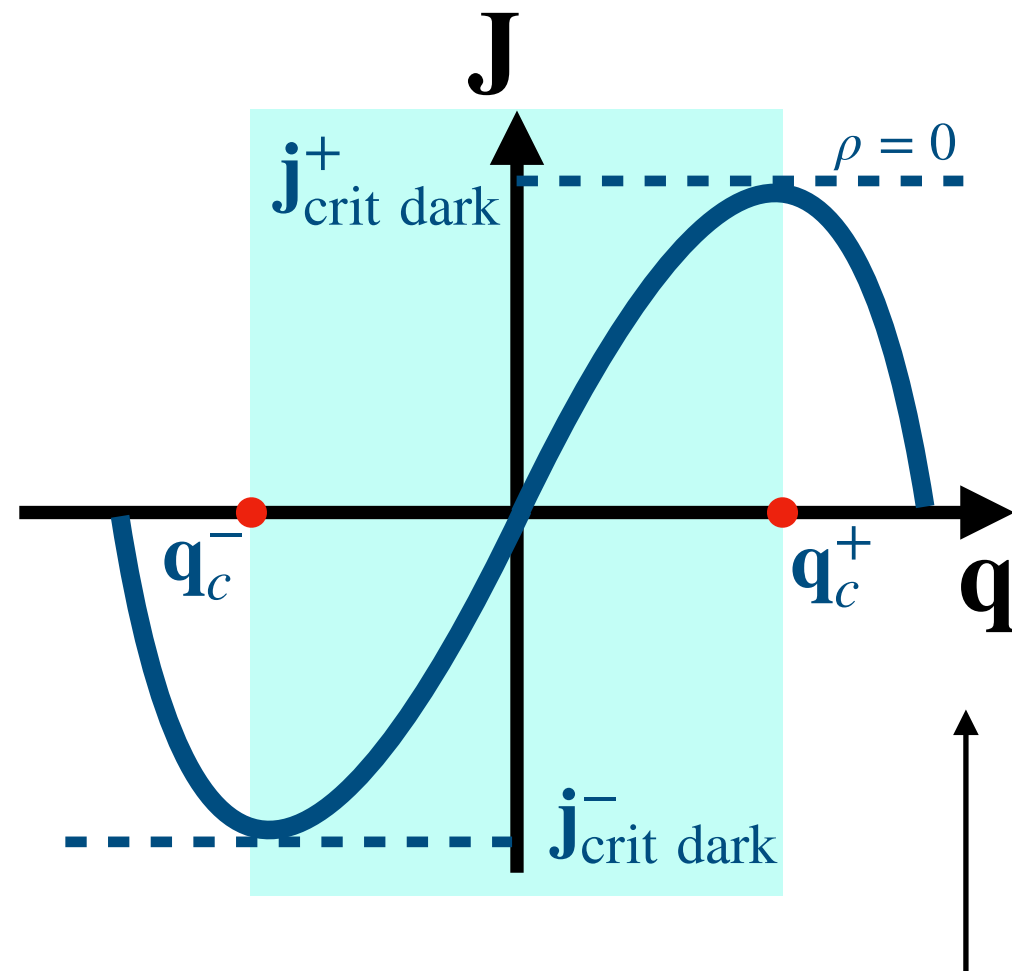
1. If a steady photocurrent flows through entire circuit there will be a potential drop across external circuit in parallel with superconductor

2. Superconductor cannot sustain finite voltage drops

3. System will adjust to zero DC voltage drop and resulting in zero photocurrent

QN: How to measure rectified photocurrent in superconductor?

Current vs momentum in superconductor



change of current is driven by
change of momentum

$$\delta \mathbf{J} = \frac{e}{\hbar} \rho_s(\mathbf{q}) \delta \mathbf{q}$$

dynamical current

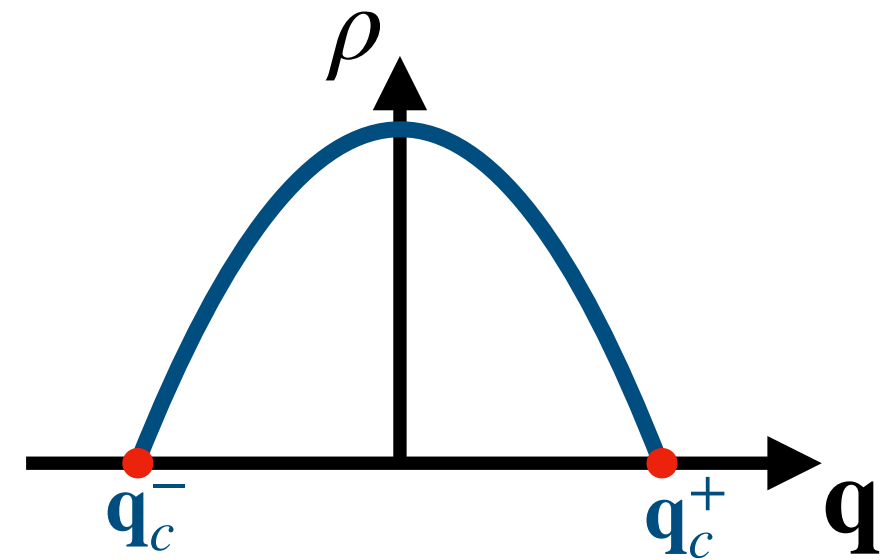
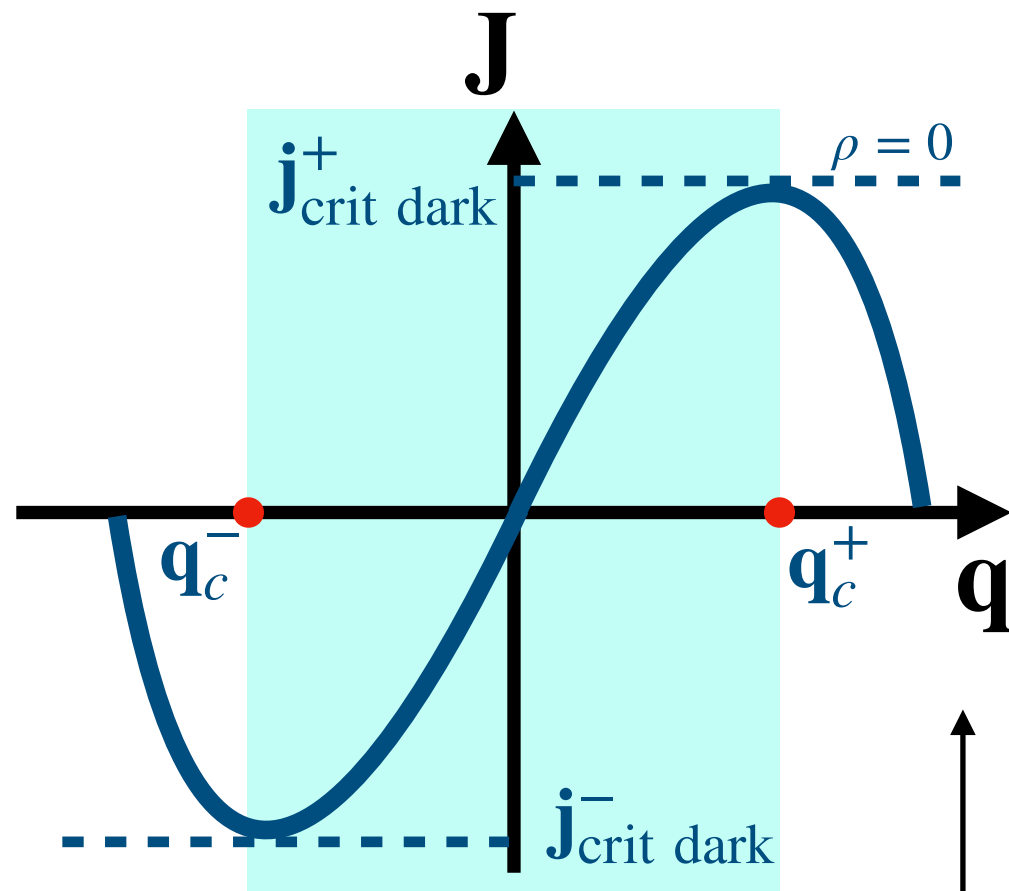
$$\dot{\mathbf{q}} = e \mathbf{E}(t)$$

$$\text{Im} \sigma(\omega) = \frac{e^2}{\hbar} \frac{\rho_s(\mathbf{q})}{\hbar \omega}$$

center of mass
momentum of cooper
pairs

Current vs momentum in superconductor

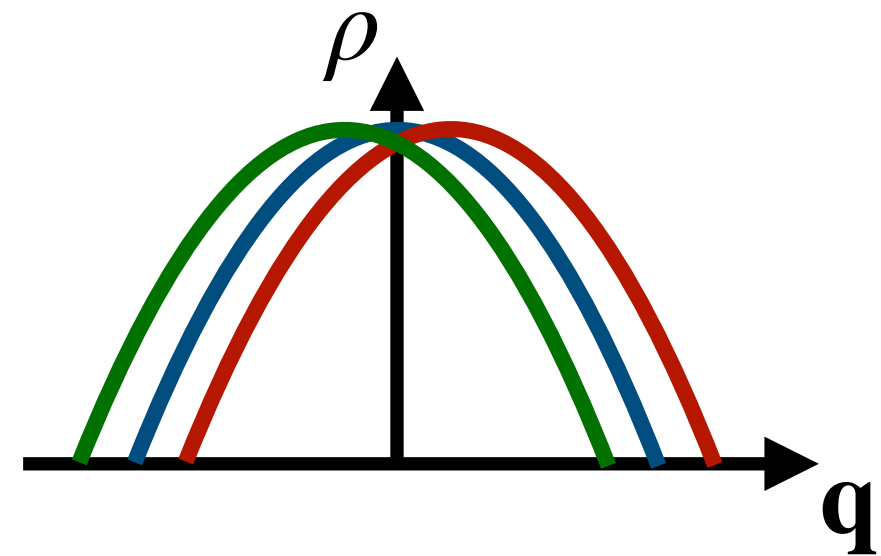
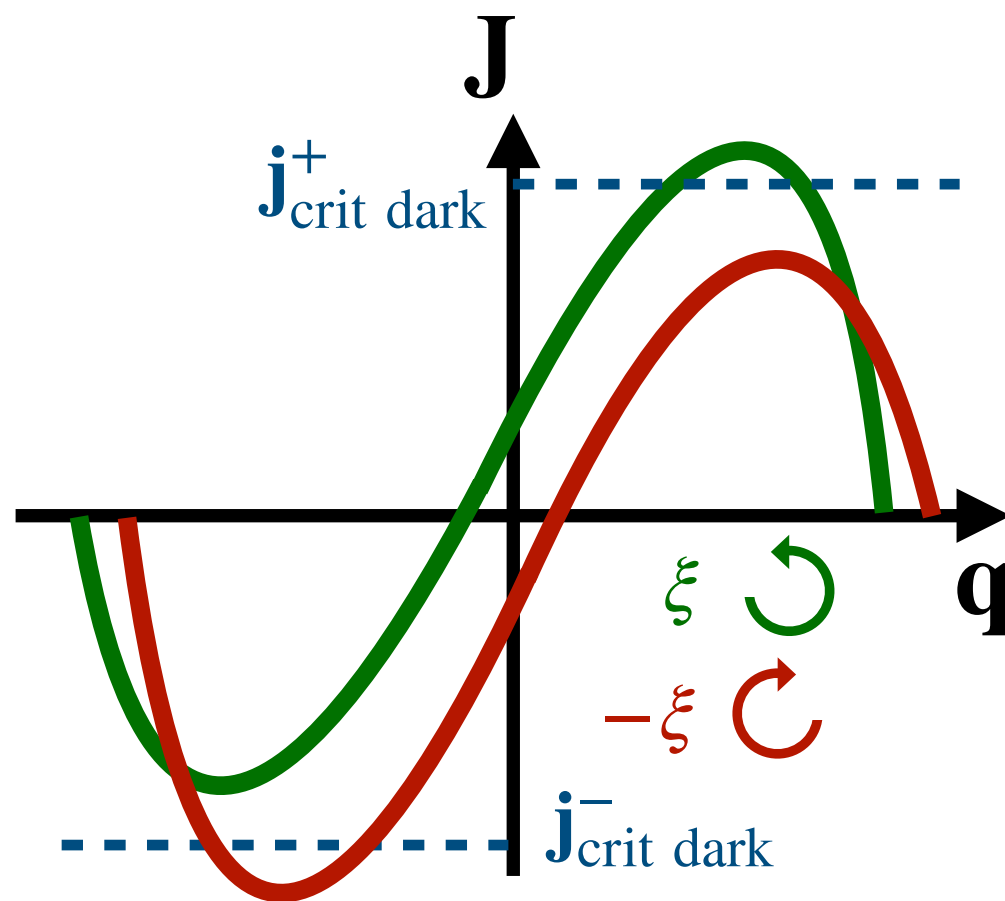
Current vs momentum in the “Dark”



center of mass
momentum of cooper
pairs

Current vs momentum in superconductor

Current vs momentum in the “light”



Current vs momentum in superconductor

adiabatic evolution of current

$$\mathbf{j}(\mathbf{q}, t) = \frac{e}{\hbar} \int_{\mathbf{k}} \sum_a f_a (\partial_{\mathbf{q}} \varepsilon_{\mathbf{k},a}(\mathbf{q}) - \hbar \dot{\mathbf{q}} \times [\nabla_{\mathbf{q}} \times \mathcal{A}_{\mathbf{k},aa}(\mathbf{q})]),$$

superconducting Berry curvature dipole

$$\mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma}[\mathbf{q}] = \int_{\mathbf{k}} \sum_a f_a \partial_{\mathbf{q}}^{\alpha} [\partial_{\mathbf{q}}^{\beta} \mathcal{A}_{\mathbf{k},aa}^{\gamma}(\mathbf{q}) - \partial_{\mathbf{q}}^{\gamma} \mathcal{A}_{\mathbf{k},aa}^{\beta}(\mathbf{q})]$$

Photocurrent underneath the gap

$$j_{\text{photo}}^{\gamma}[\mathbf{q}] = \frac{2e^3}{\hbar} \left\{ \frac{\varepsilon^{*\alpha} \varepsilon^{\beta}}{\hbar^2 \omega_d^2} \mathcal{J}_{\mathbf{q}}^{\alpha\beta\gamma} + \frac{\text{Im}[\varepsilon^{*\alpha} \varepsilon^{\beta}]}{\hbar \omega_d} \mathcal{D}_{\mathbf{q}}^{\alpha\beta\gamma} \right\},$$

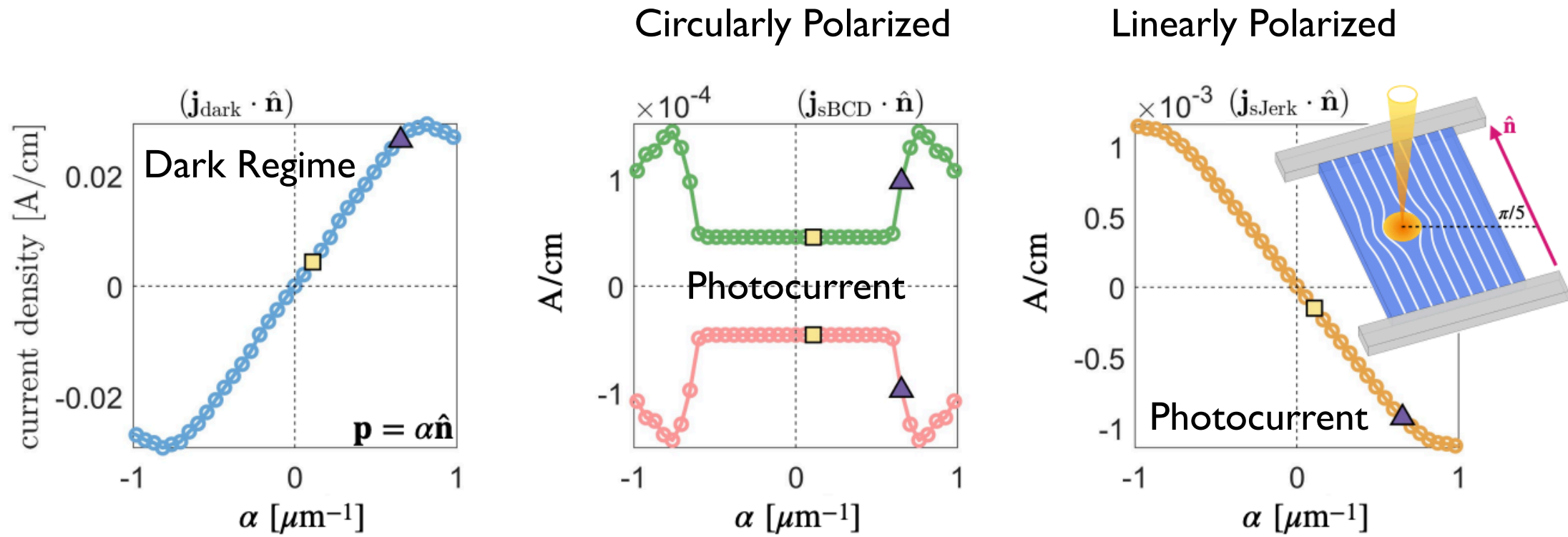
dependence on $\mathbf{q}$ in TRS preserving ground state

$$\mathcal{J}_{\mathbf{q}}^{\alpha\beta\gamma} = \int_{\mathbf{k}} \sum_a f_a \partial_{\mathbf{q}}^{\gamma} \partial_{\mathbf{q}}^{\beta} \partial_{\mathbf{q}}^{\alpha} \varepsilon_{\mathbf{k},a}(\mathbf{q})$$

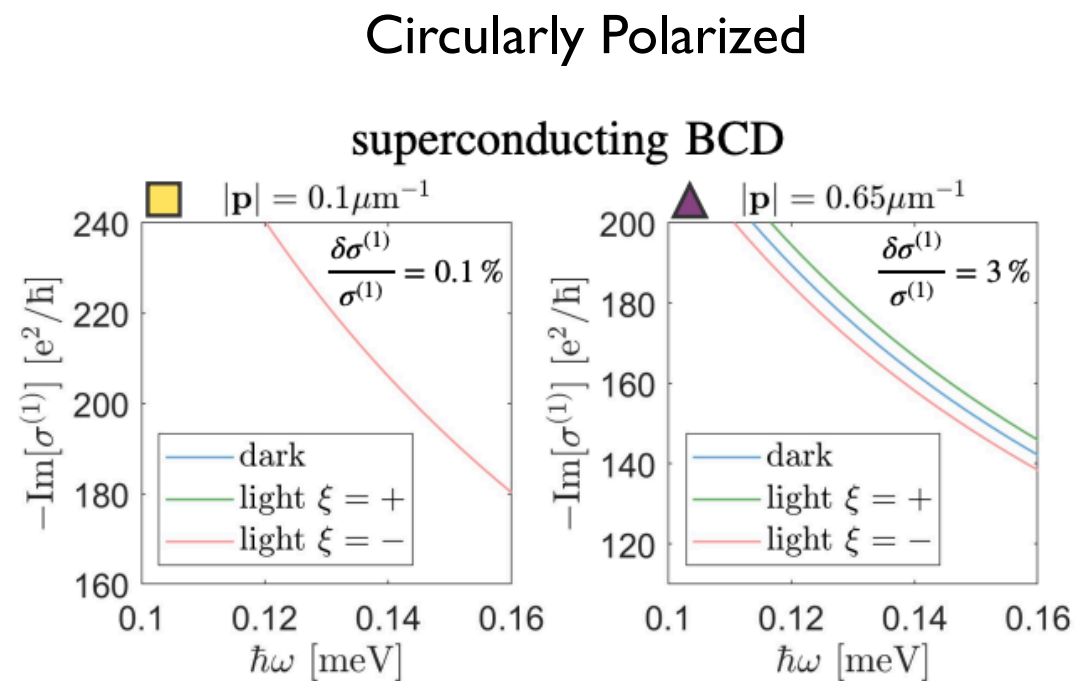
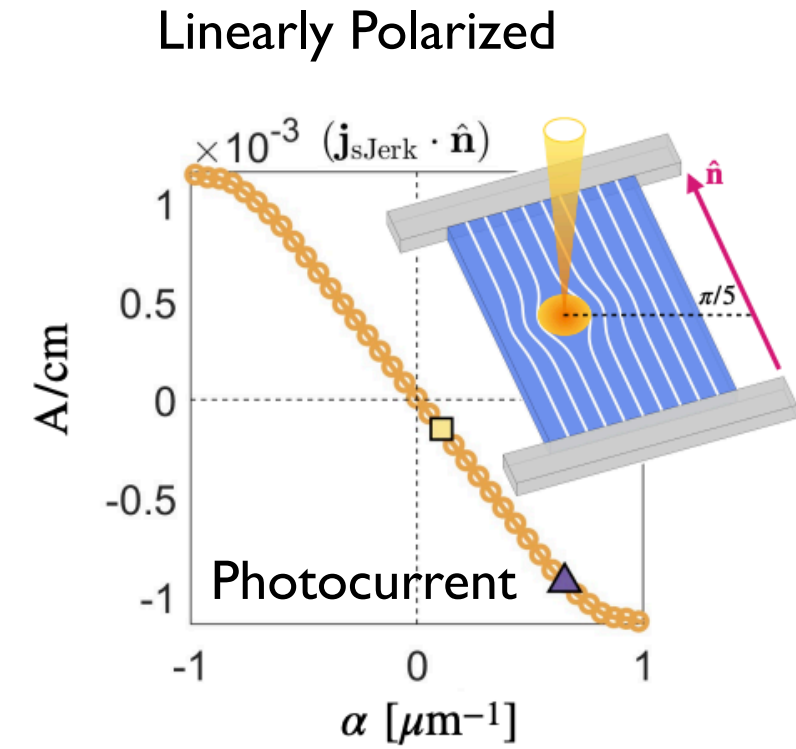
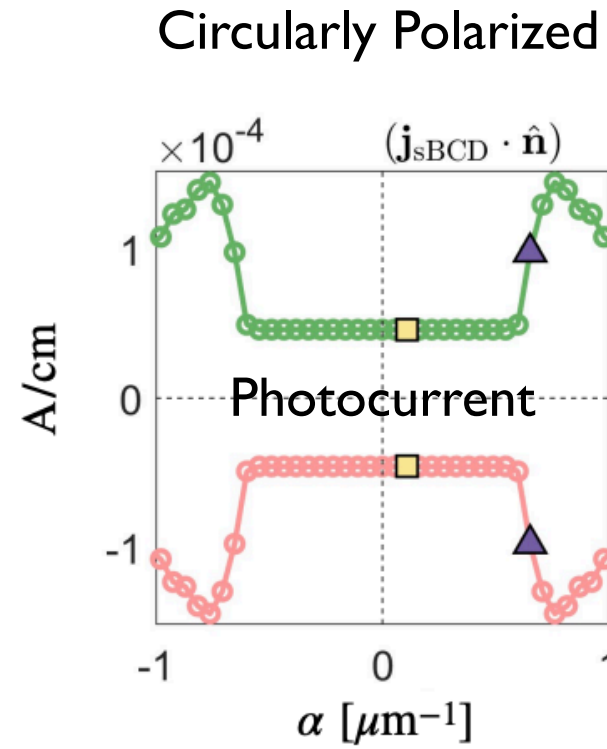
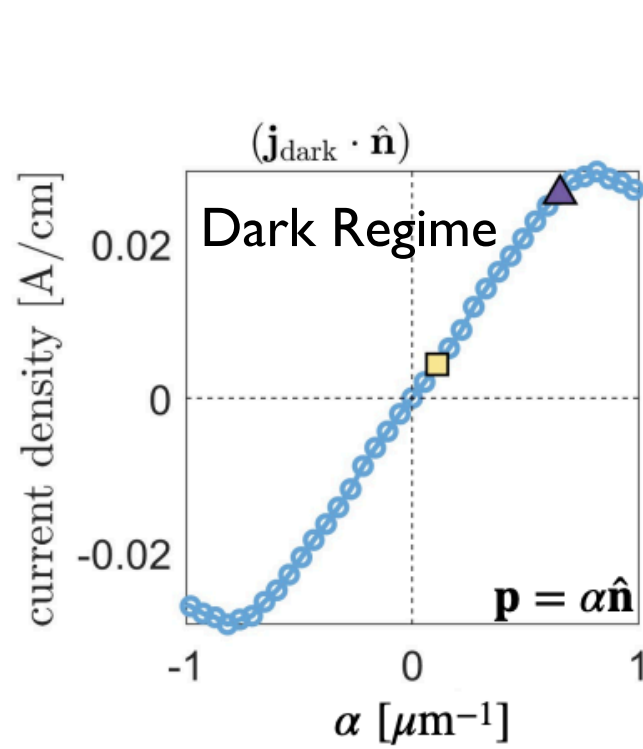
“Jerk” (second derivative of velocity)

	q-parity	Control
sBCD	+	CL
sJerk	-	LL

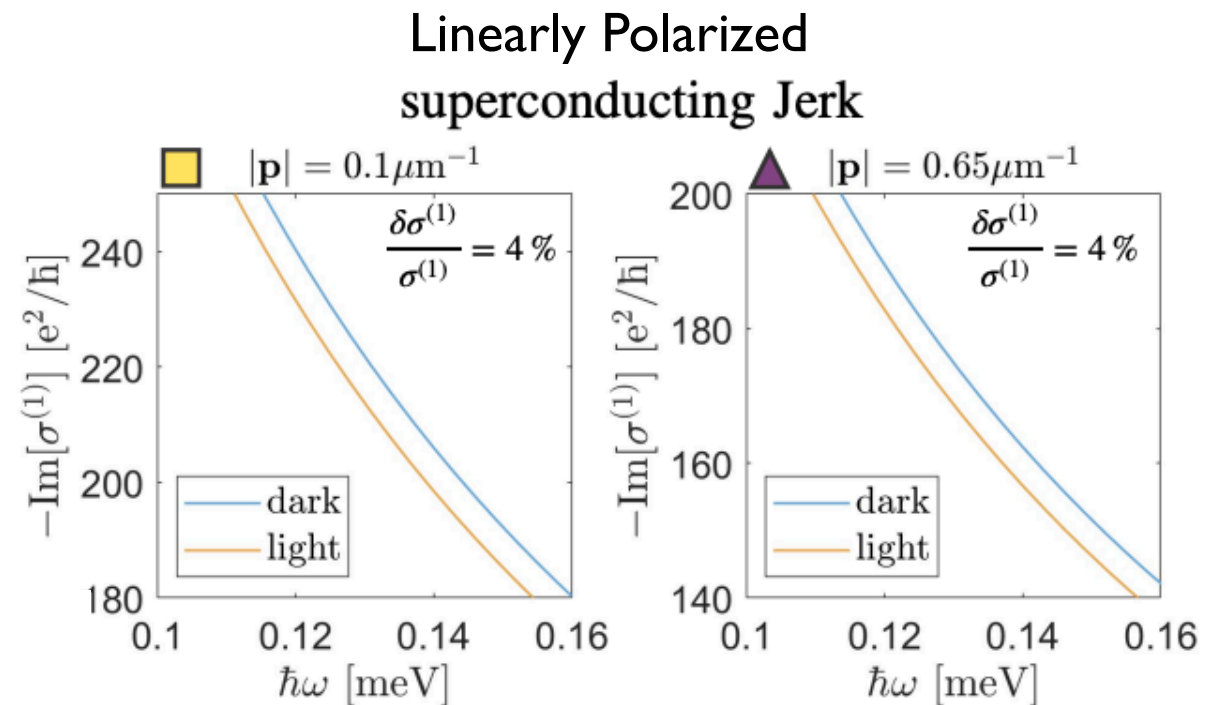
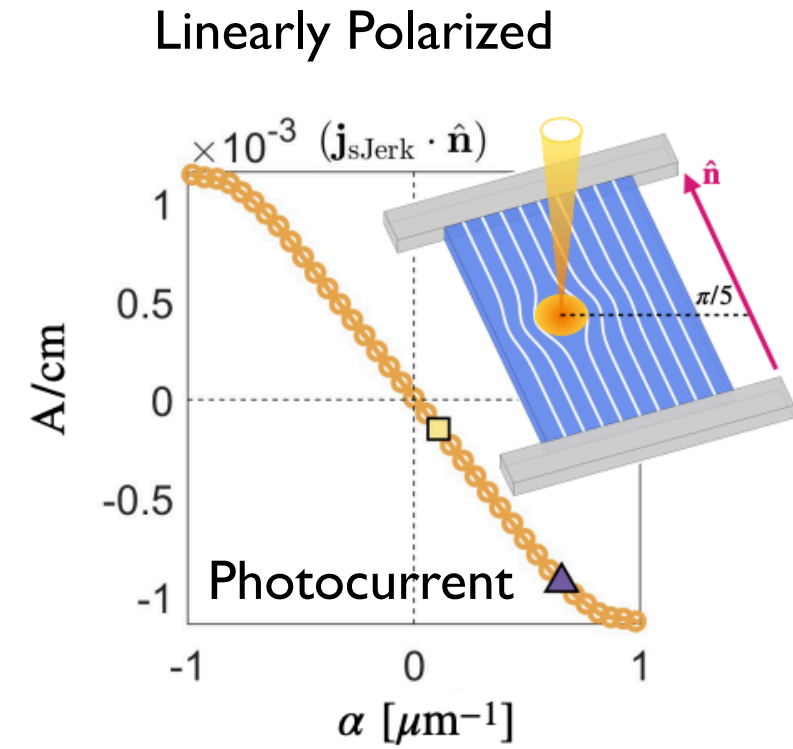
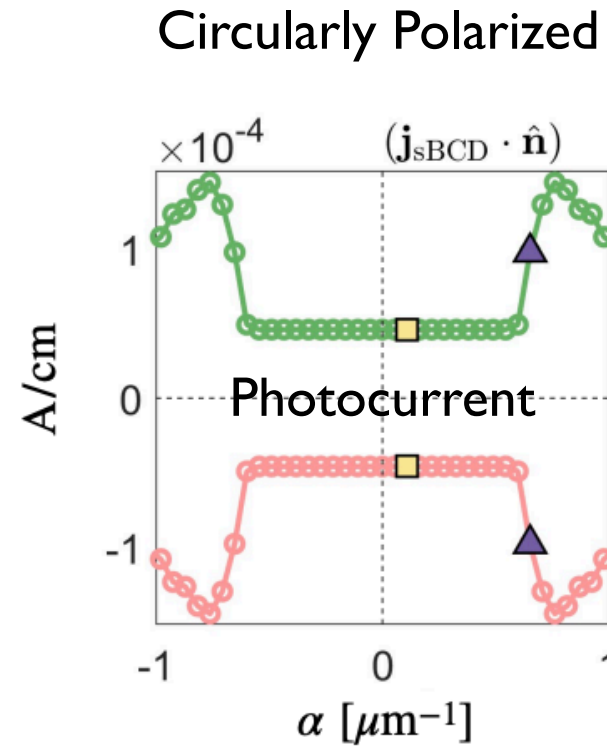
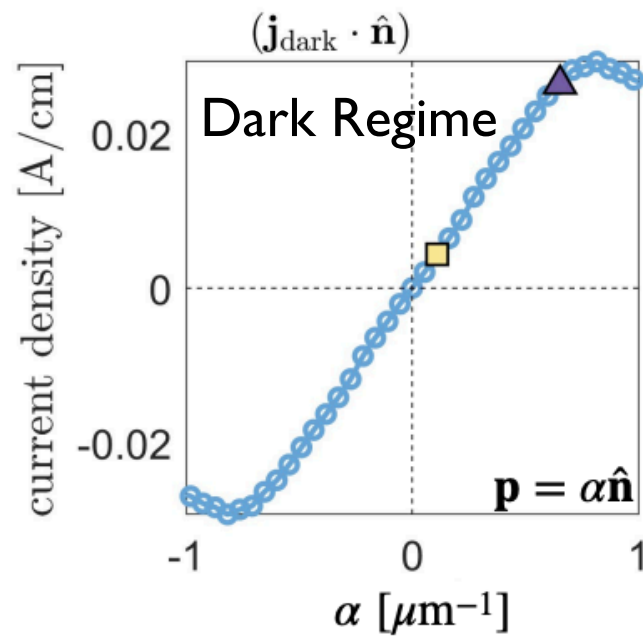
Photocurrent modifies current-momentum relation



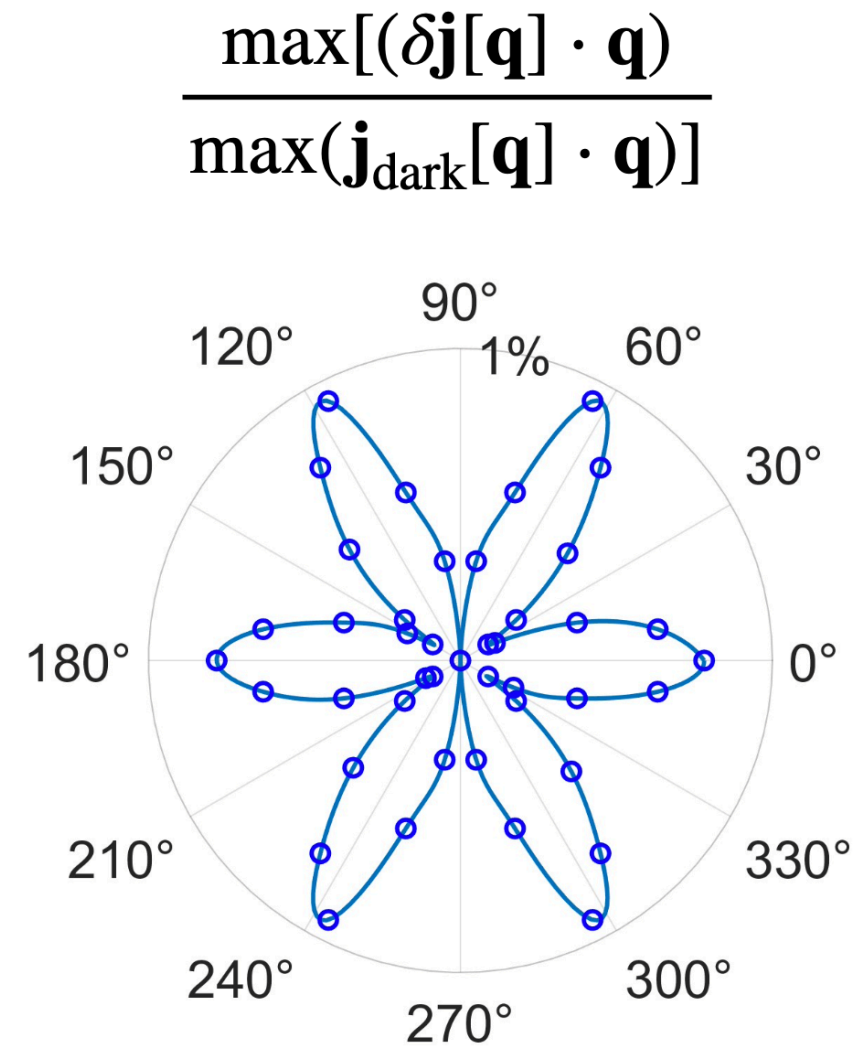
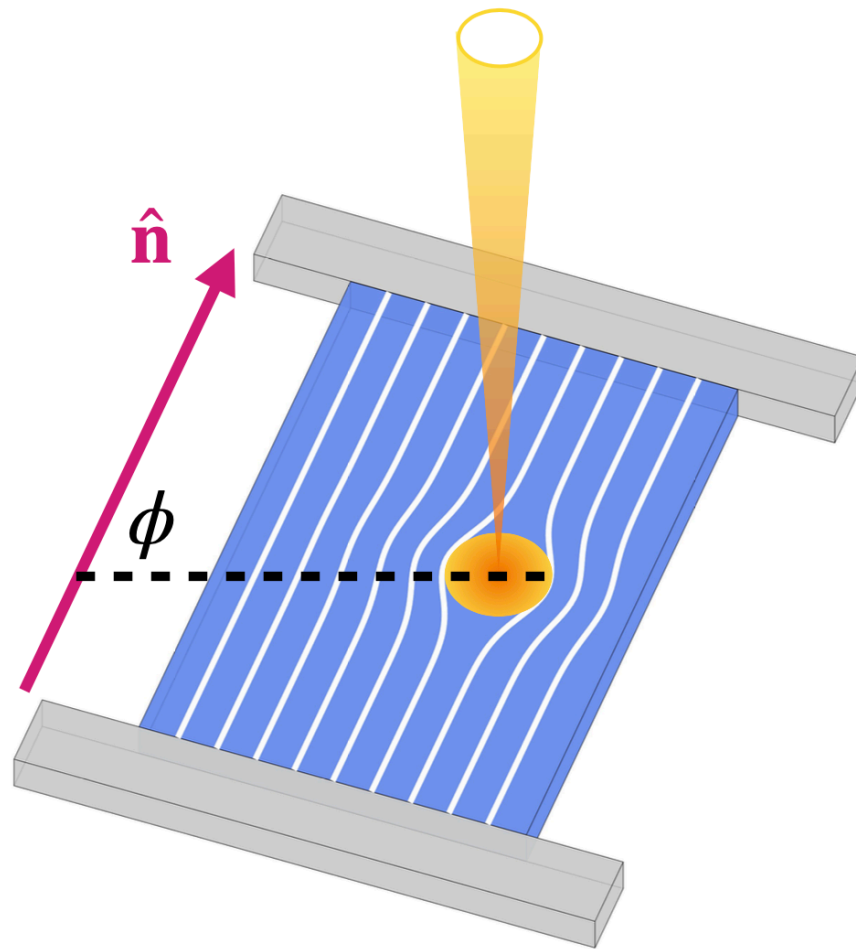
Photocurrent modifies current-momentum relation



Photocurrent modifies current-momentum relation



Light induced diode effect



Diversity of second-order nonlinear photocurrents

Real transitions between two bands

e.g.,

(i) Shift Photocurrent

arises from “shift” in the quasiparticle’s center of mass when it transitions from valence to conduction band

e.g., von Blatz and Kraut PRB (1981), Morimoto and Nagaosa, Sci. Adv (2016)....

(ii) Injection Photocurrent

arises from difference in a particle’s velocity when it transitions from valence to conduction band

e.g., Sipe PRB (2000), De Juan et al., Nat Comm (2016)....

Non-exhaustive list: Other types of bulk photocurrents can also be induced, these are just a sampling of some of the most well-known. Lots of other papers other than the ones shown here too; these are just examples.

“Metallic” photocurrents

so-called “Fermi surface” contributions

e.g.,

(iii) Berry curvature dipole nonlinear transport

arises from (berry curvature induced) anomalous velocity contributions to the semiclassical equations of motion (in a band)

e.g., TRS: Orenstein and Moore PRL (2010), Sodemann and Fu PRL (2015)
PT: Gao, Yang, Niu (2014), Wang, Gao, Xiao PRL (2020),

(iv) Extrinsic scattering

e.g., from skew scattering-type processes

e.g., TRS: Du, .., Xie Nat Comm (2019), Konig, .., Pesin PRB (2019),
Isobe, Xu, L Fu Sci Adv (2020),
PT: M Da, A Arora, G Vignale, JS PRL (2023)

(v) Intrinsic Fermi surface effect

arising from off-resonant contributions in the second order density matrix

e.g., de Juan, Grushin PRR (2020), Wantanabe and Yanase PRX (2021), Gao, ... Rappe PRR (2021),

BCD is a fundamental characteristic of Bloch electrons

velocity of Bloch electron

$$\hat{v}_n^\alpha = \frac{i}{\hbar} [\hat{H}(\mathbf{k}), \hat{r}^\alpha]_n = \frac{1}{\hbar} \frac{\partial}{\partial k^\alpha} \epsilon_n - e \frac{E^\beta(t)}{\hbar} \hat{\Omega}_n^{\beta\alpha}.$$

Berry curvature dipole gives
nonlinear acceleration

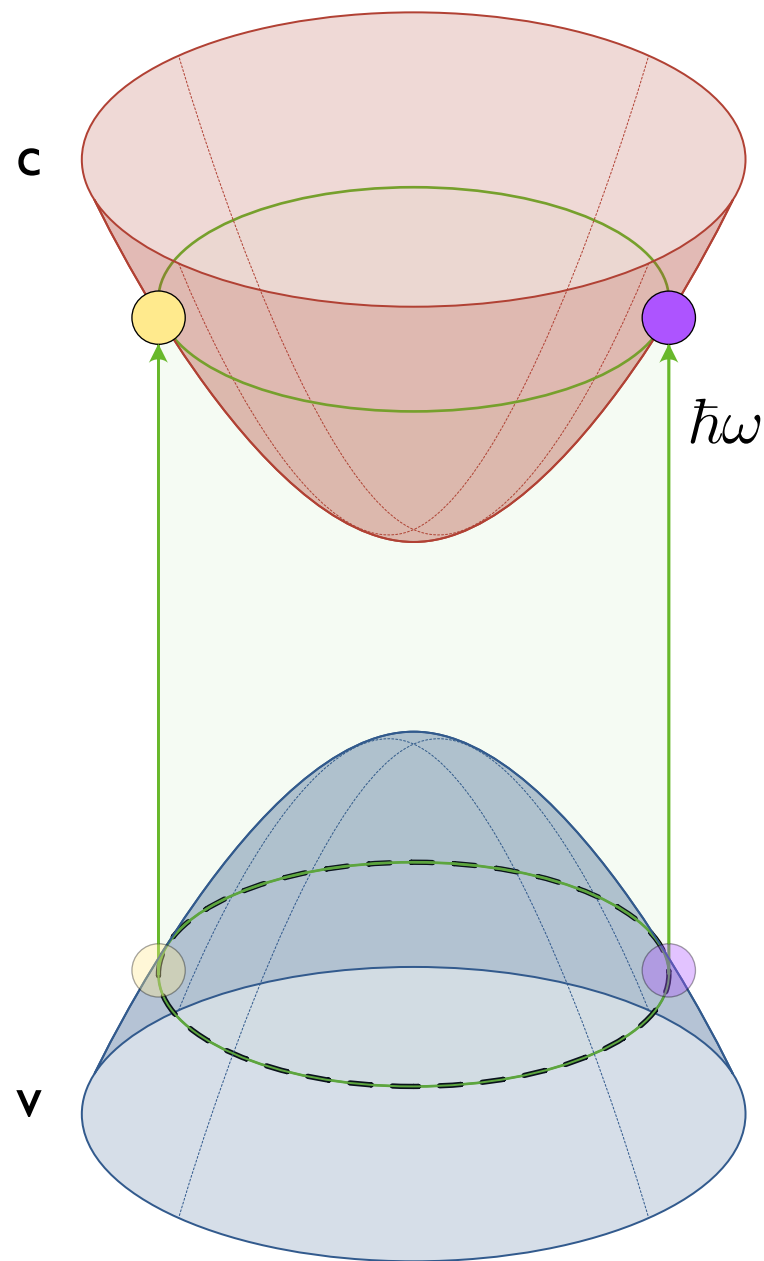
acceleration of Bloch electron:

$$a^\gamma = -e \frac{E^\alpha}{\hbar^2} \frac{\partial^2 \epsilon}{\partial k^\alpha \partial k^\gamma} + e^2 \frac{E^\alpha E^\beta}{\hbar^2} \frac{\partial}{\partial k^\alpha} \Omega^{\beta\gamma}$$

inverse effective mass

Matsyshyn, Sodemann PRL (2019)

Light-induced change in velocity: *injection PC* + *Ballistic BC*



Injection photocurrent:

Interband transition rate



$$\partial_t \mathbf{j}^{\text{inj}} = e \sum_{i \rightarrow f} W_{i \rightarrow f} \Delta_{i \rightarrow f},$$

Difference in velocity: $\Delta_{i \rightarrow f} = \mathbf{v}_f - \mathbf{v}_i,$

injection current: normal Fermi golden rule type W

Ballistic current: dressed W (e.g., to allow an imbalance of rates even for k and $-k$ in a TRS material)

Order parameter BCD

BdG hamiltonian:

$$\hat{H}_{\mathbf{k},\mathbf{q}}^{\text{BdG}}[\mathbf{A}(t)] = \begin{pmatrix} \hat{H}_{\mathbf{k}+\mathbf{q}}^{(0)}[\mathbf{A}(t)] & \hat{\Delta} \\ \hat{\Delta}^\dagger & -\{\hat{H}_{-\mathbf{k}+\mathbf{q}}^{(0)}[\mathbf{A}(t)]\}^T \end{pmatrix}$$

Depends on phase of gap parameter

$$\mathbf{L}_n(\mathbf{k}, \mathbf{q}) = -\partial_{\mathbf{q}} \arg \left[\frac{\langle n, \mathbf{k} + \mathbf{q} | \hat{\Delta} | n, \mathbf{k} - \mathbf{q} \rangle}{\langle n, \mathbf{k} + \mathbf{q} | n, \mathbf{k} - \mathbf{q} \rangle} \right].$$

$$\begin{aligned} \mathcal{D}_{\text{sBCD}}^{\alpha\beta\gamma} = & \int_{\mathbf{k}} \underbrace{\left[L_n^\gamma(\mathbf{k}, \mathbf{q}) \partial_{\mathbf{q}}^\alpha \partial_{\mathbf{q}}^\beta |v_{\mathbf{k},\mathbf{q}}|^2 - (\beta \leftrightarrow \gamma) \right]}_{\text{order parameter BCD distribution}} \Big|_{\mathbf{q} \rightarrow 0} \\ & + 2 \int_{\mathbf{k}} |v_{\mathbf{k},0}|^2 \underbrace{(\partial_{\mathbf{k}}^\alpha \partial_{\mathbf{k}}^\beta r_n^\gamma(\mathbf{k}) - \partial_{\mathbf{k}}^\alpha \partial_{\mathbf{k}}^\gamma r_n^\beta(\mathbf{k}))}_{\text{parent BCD distribution}}, \end{aligned}$$

Bloch state BCD distribution

